**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

DAIRY FACILITY INFORMATION

**A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY:** Joe Oliveira Dairy

Physical address of dairy:

| Number and Street | City | County | Zip Code |
|---|---|---|---|
| 19104 Crane AVE | Hilmar | Merced | 95324 |

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 12/02/1986

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X220-X012-XXXX

**B. OPERATORS**

Oliveira, Joe

Operator name: Oliveira, Joe

Telephone no.: (209) 652-5397 (Landline) (Cellular)

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 19104 Crane AVE | Hilmar | CA | 95324 |

**This operator is responsible for paying permit fees.**

**C. OWNERS**

Oliveira, Joe

Legal owner name: Oliveira, Joe

Telephone no.: (209) 652-5397 (Landline) (Cellular)

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 19104 Crane AVE | Hilmar | CA | 95324 |

**This owner is responsible for paying permit fees.**

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AVAILABLE NUTRIENTS

## A. HERD INFORMATION

| | Milk Cows | Dry Cows | Bred Heifers (15-24 mo.) | Heifers (7-14 mo. to breeding) | Calves (4-6 mo.) | Calves (0-3 mo.) |
|---|---|---|---|---|---|---|
| Number open confinement | 0 | 0 | 445 | 420 | 321 | 0 |
| Number under roof | 1,200 | 54 | 0 | 0 | 0 | 200 |
| Maximum number | 1,200 | 54 | 445 | 420 | 321 | 200 |
| Average number | 1,200 | 54 | 445 | 420 | 321 | 200 |
| Avg live weight (lbs) | 1,500 | 1,550 | 950 | 500 | | |

Predominant milk cow breed: Holstein

Average milk production: 80 *pounds per cow per day*

## B. MANURE GENERATED

Total manure excreted by the herd: 43,095.33 *tons per reporting period*

Total nitrogen from manure: 539,436.24 *lbs per reporting period*

After ammonia losses (30% loss applied): 377,605.37 *lbs per reporting period*

Total phosphorus from manure: 88,594.53 *lbs per reporting period*

Total potassium from manure: 237,695.07 *lbs per reporting period*

Total salt from manure: 577,437.30 *lbs per reporting period*

## C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 13,200,294 *gallons*

Total nitrogen generated: 80,170.71 *lbs*

Total phosphorus generated: 9,484.37 *lbs*

Total potassium generated: 97,598.41 *lbs*

Total salt generated: 611,361.69 *lbs*

| | | |
|---|---|---|
| | 22,992,294 | *gallons applied* |
| + | 0 | *gallons exported* |
| - | 9,792,000 | *gallons imported* |
| = | 13,200,294 | *gallons generated* |

## D. FRESH WATER SOURCES

| Source Description | Type |
|---|---|
| Domestic Well - Domestic Well East | Ground water |
| Domestic Well - Domestic Well West | Ground water |
| Irrigation Pump | Ground water |
| Irrigation Water - Canal | Surface water |

## E. SUBSURFACE (TILE) DRAINAGE SOURCES

*No subsurface (tile) drainage sources entered.*

## F. NUTRIENT IMPORTS

*No dry manure nutrient imports entered.*

| Date | Material type / Description | Quantity | N (mg/L) | P (mg/L) | K (mg/L) | Salt (mg/L) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|
| 02/08/2023 | Process wastewater<br>*LWI - HH* | 2,592,000.00 *gal* | 258.00 | 45.50 | 805.00 | | 2,880 |
| 05/30/2023 | Process wastewater<br>*LWI - HH* | 3,456,000.00 *gal* | 608.00 | 35.30 | 985.00 | | 6,060 |
| 08/05/2023 | Process wastewater<br>*LWI - HH* | 3,744,000.00 *gal* | 493.00 | 76.40 | 800.00 | | 3,660 |

*No commercial or other nutrient imports entered.*

| Material type | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 38,518.65 | 4,389.26 | 70,815.00 | 351,419.30 |
| Total imports for all materials | 38,518.65 | 4,389.26 | 70,815.00 | 351,419.30 |

## G. NUTRIENT EXPORTS

*No solid nutrient exports entered.*

*No liquid nutrient exports entered.*

APPLICATION AREA

## A. LIST OF LAND APPLICATION AREAS

| Field name | Controlled acres | Cropable acres | Total harvests | Type of waste applied | Parcel number |
|---|---|---|---|---|---|
| Field 1 | 40 | 40 | 2 | process wastewater | X045-X220-X012-XXXX |
| Field 2 | 10 | 10 | 2 | process wastewater | X045-X220-X062-XXXX |
| Field 3 | 40 | 40 | 2 | process wastewater | X045-X220-X062-XXXX |
| Field 4 | 45 | 45 | 2 | process wastewater | X045-X220-X058-XXXX |
| Field 5 | 5 | 5 | 2 | process wastewater | X045-X220-X062-XXXX |
| Field ME | 27 | 27 | 2 | both | X045-X240-X035-XXXX |
| Field MW | 38 | 38 | 2 | both | X045-X240-X035-XXXX |
| Field NE | 35 | 35 | 2 | both | X045-X240-X035-XXXX |
| Field NW | 38 | 38 | 2 | both | X045-X240-X035-XXXX |
| Field SE | 34 | 34 | 2 | both | X045-X250-X082-XXXX |
| Field SW | 36 | 36 | 2 | both | X045-X251-X001-XXXX |
| Totals for areas that were used for application | 348 | 348 | 22 | | |
| Totals for areas that were not used for application | | | | | |
| Land application area totals | 348 | 348 | 22 | | |

## B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/19/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/29/2023 | 402.60 *ton* | Dry-weight | | 65.1 | 29,000.00 | 5,500.00 | 44,500.00 | | 13.80 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 10.07 | 203.74 | 38.64 | 312.63 | 969.50 |

## Field 1

### 05/21/2023: Corn, silage

Crop: Corn, silage  Acres planted: 40  Plant date: 05/21/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/08/2023 | 1,145.60 *ton* | Dry-weight | | 72.8 | 15,500.00 | 3,100.00 | 19,300.00 | | 6.80 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 28.64 | 241.49 | 48.30 | 300.70 | 1,059.45 |

## Field 2

Field name: Field 2

### 11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough  Acres planted: 10  Plant date: 11/19/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/29/2023 | 98.90 *ton* | Dry-weight | | 64.8 | 30,200.00 | 5,200.00 | 43,200.00 | | 14.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 9.89 | 210.27 | 36.21 | 300.78 | 1,002.61 |

### 05/21/2023: Corn, silage

Crop: Corn, silage  Acres planted: 10  Plant date: 05/21/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/08/2023 | 308.70 *ton* | Dry-weight | | 72.0 | 15,100.00 | 3,200.00 | 18,800.00 | | 6.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 30.87 | 261.04 | 55.32 | 325.00 | 1,106.38 |

## Field 3

Field name: Field 3

### 11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/19/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/29/2023 | 360.20 *ton* | Dry-weight | | 63.0 | 30,400.00 | 5,500.00 | 43,600.00 | | 14.20 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 9.01 | 202.58 | 36.65 | 290.54 | 946.25 |

### 05/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/21/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/08/2023 | 1,122.30 *ton* | Dry-weight | | 70.1 | 15,000.00 | 3,200.00 | 19,400.00 | | 6.90 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 28.06 | 251.68 | 53.69 | 325.50 | 1,157.71 |

## Field 4

Field name: Field 4

### 11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 45 Plant date: 11/19/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/29/2023 | 405.30 *ton* | Dry-weight | | 66.1 | 30,600.00 | 5,500.00 | 42,400.00 | | 18.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 9.01 | 186.86 | 33.59 | 258.92 | 1,099.17 |

## Field 4

### 05/21/2023: Corn, silage

Crop: Corn, silage  Acres planted: 45  Plant date: 05/21/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/08/2023 | 1,173.00 *ton* | Dry-weight | | 70.4 | 18,000.00 | 3,100.00 | 19,600.00 | | 7.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 26.07 | 277.77 | 47.84 | 302.46 | 1,080.20 |

## Field 5

Field name: Field 5

### 11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough  Acres planted: 5  Plant date: 11/19/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/29/2023 | 45.10 *ton* | Dry-weight | | 64.8 | 30,200.00 | 5,200.00 | 43,200.00 | | 14.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 9.02 | 191.77 | 33.02 | 274.32 | 914.41 |

### 05/21/2023: Corn, silage

Crop: Corn, silage  Acres planted: 5  Plant date: 05/21/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/08/2023 | 143.20 *ton* | Dry-weight | | 72.0 | 15,100.00 | 3,200.00 | 18,800.00 | | 6.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 28.64 | 242.18 | 51.32 | 301.52 | 1,026.46 |

## Field ME

Field name: Field ME

### 11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 27 Plant date: 11/23/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/19/2023 | 378.60 *ton* | Dry-weight | | 66.5 | 17,500.00 | 3,700.00 | 2,600.00 | | 7.70 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 176.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 14.02 | 164.41 | 34.76 | 24.43 | 723.41 |

### 06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 06/14/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/12/2023 | 712.50 *ton* | Dry-weight | | 58.7 | 11,200.00 | 2,300.00 | 15,300.00 | | 4.80 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 26.39 | 244.13 | 50.13 | 333.50 | 1,046.27 |

## Field MW

Field name: Field MW

### 11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 11/23/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/19/2023 | 532.60 *ton* | Dry-weight | | 66.8 | 14,900.00 | 3,300.00 | 24,700.00 | | 7.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 14.02 | 138.67 | 30.71 | 229.87 | 688.68 |

## Field MW

### 06/14/2023: Corn, silage

Crop: Corn, silage  Acres planted: 38  Plant date: 06/14/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/12/2023 | 1,002.30 *ton* | Dry-weight | | 60.8 | 12,100.00 | 2,500.00 | 15,300.00 | | 4.90 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 26.38 | 250.22 | 51.70 | 316.39 | 1,013.27 |

## Field NE

Field name: Field NE

### 11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough  Acres planted: 35  Plant date: 11/23/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/19/2023 | 475.60 *ton* | Dry-weight | | 66.8 | 17,300.00 | 3,500.00 | 27,900.00 | | 7.80 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 13.59 | 156.09 | 31.58 | 251.74 | 703.78 |

### 06/14/2023: Corn, silage

Crop: Corn, silage  Acres planted: 35  Plant date: 06/14/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/12/2023 | 912.60 *ton* | Dry-weight | | 59.2 | 11,800.00 | 2,800.00 | 15,200.00 | | 4.90 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 26.07 | 251.06 | 59.57 | 323.40 | 1,042.55 |

## Field NW

Field name: Field NW

### 11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 11/23/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/19/2023 | 556.60 *ton* | Dry-weight | | 67.6 | 15,400.00 | 3,500.00 | 27,300.00 | | 7.30 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 14.65 | 146.17 | 33.22 | 259.12 | 692.88 |

### 06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/14/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/12/2023 | 1,023.60 *ton* | Dry-weight | | 60.6 | 11,400.00 | 2,300.00 | 15,500.00 | | 4.80 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 26.94 | 241.98 | 48.82 | 329.01 | 1,018.86 |

## Field SE

Field name: Field SE

### 11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 34 Plant date: 11/23/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/19/2023 | 378.50 *ton* | Dry-weight | | 64.9 | 21,200.00 | 4,000.00 | 31,900.00 | | 8.70 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 11.13 | 165.68 | 31.26 | 249.30 | 679.90 |

## Field SE

### 06/14/2023: Corn, silage

Crop: Corn, silage  Acres planted: 34  Plant date: 06/14/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/12/2023 | 983.40 *ton* | Dry-weight | | 60.8 | 10,700.00 | 2,000.00 | 14,900.00 | | 4.80 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 28.92 | 242.63 | 45.35 | 337.87 | 1,088.45 |

## Field SW

Field name: Field SW

### 11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough  Acres planted: 36  Plant date: 11/23/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/19/2023 | 412.30 *ton* | Dry-weight | | 67.2 | 19,800.00 | 3,800.00 | 30,700.00 | | 7.50 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 11.45 | 148.76 | 28.55 | 230.65 | 563.48 |

### 06/14/2023: Corn, silage

Crop: Corn, silage  Acres planted: 35  Plant date: 06/14/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/12/2023 | 1,012.50 *ton* | Dry-weight | | 60.5 | 10,800.00 | 2,100.00 | 17,900.00 | | 5.50 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 28.93 | 246.82 | 47.99 | 409.08 | 1,256.95 |

NUTRIENT BUDGET

## A. LAND APPLICATIONS

### Field 1 - 11/19/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 11/19/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 127.54 | 15.09 | 155.27 | 972.61 | 840,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.93 | 4,916,550.00 *gal* |
| Application event totals | | 127.54 | 15.09 | 155.27 | 1,009.54 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 120.26 | 14.23 | 146.39 | 917.03 | 792,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 34.14 | 4,546,125.00 *gal* |
| Application event totals | | 120.26 | 14.23 | 146.39 | 951.18 | |

### Field 1 - 05/21/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 05/21/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 45.27 | 6,027,825.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 45.27 | |

Field 1 - 05/21/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/07/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 88.37 | 10.45 | 107.58 | 673.88 | 582,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 45.78 | 6,095,175.00 *gal* |
| Application event totals | | 88.37 | 10.45 | 107.58 | 719.66 | |

| | | | | |
|---|---|---|---|---|
| 06/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 48.81 | 6,499,275.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 48.81 | |

| | | | | |
|---|---|---|---|---|
| 07/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 81.99 | 9.70 | 99.81 | 625.25 | 540,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 46.28 | 6,162,525.00 *gal* |
| Application event totals | | 81.99 | 9.70 | 99.81 | 671.53 | |

| | | | | |
|---|---|---|---|---|
| 07/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 81.99 | 9.70 | 99.81 | 625.25 | 540,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 43.50 | 5,792,100.00 *gal* |
| Application event totals | | 81.99 | 9.70 | 99.81 | 668.75 | |

| | | | | |
|---|---|---|---|---|
| 08/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 76.53 | 9.05 | 93.16 | 583.57 | 504,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 45.78 | 6,095,175.00 *gal* |
| Application event totals | | 76.53 | 9.05 | 93.16 | 629.34 | |

## Field 1 - 05/21/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 43.25 | 5,758,425.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 43.25 | |

## Field 2 - 11/19/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough Plant date: 11/19/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 131.19 | 15.52 | 159.70 | 1,000.40 | 216,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 51.59 | 1,717,425.00 *gal* |
| Application event totals | | 131.19 | 15.52 | 159.70 | 1,051.99 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 123.90 | 14.66 | 150.83 | 944.82 | 204,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.45 | 1,313,325.00 *gal* |
| Application event totals | | 123.90 | 14.66 | 150.83 | 984.28 | |

## Field 2 - 05/21/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage Plant date: 05/21/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|

Field 2 - 05/21/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 30.35 | 1,010,250.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 30.35 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/07/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 87.46 | 10.35 | 106.47 | 666.93 | 144,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.45 | 1,313,325.00 *gal* |
| Application event totals | | 87.46 | 10.35 | 106.47 | 706.39 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 41.48 | 1,380,675.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 41.48 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 87.46 | 10.35 | 106.47 | 666.93 | 144,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 41.48 | 1,380,675.00 *gal* |
| Application event totals | | 87.46 | 10.35 | 106.47 | 708.41 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 87.46 | 10.35 | 106.47 | 666.93 | 144,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 44.51 | 1,481,700.00 *gal* |
| Application event totals | | 87.46 | 10.35 | 106.47 | 711.45 | |

## Field 2 - 05/21/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 87.46 | 10.35 | 106.47 | 666.93 | 144,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 51.59 | 1,717,425.00 *gal* |
| Application event totals | | 87.46 | 10.35 | 106.47 | 718.53 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 53.62 | 1,784,775.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 53.62 | |

## Field 3 - 11/19/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 11/19/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 116.61 | 13.80 | 141.96 | 889.24 | 768,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 45.52 | 6,061,500.00 *gal* |
| Application event totals | | 116.61 | 13.80 | 141.96 | 934.77 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/07/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 100.21 | 11.86 | 122.00 | 764.19 | 660,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 43.25 | 5,758,425.00 *gal* |
| Application event totals | | 100.21 | 11.86 | 122.00 | 807.44 | |

## Field 3 - 05/21/2023: Corn, silage

## Field 3 - 05/21/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/21/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 42.74 | 5,691,075.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 42.74 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 90.19 | 10.67 | 109.80 | 687.77 | 594,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 40.72 | 5,421,675.00 *gal* |
| Application event totals | | 90.19 | 10.67 | 109.80 | 728.49 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.71 | 5,286,975.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 39.71 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 88.37 | 10.45 | 107.58 | 673.88 | 582,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 43.25 | 5,758,425.00 *gal* |
| Application event totals | | 88.37 | 10.45 | 107.58 | 717.13 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/21/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 81.99 | 9.70 | 99.81 | 625.25 | 540,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 43.25 | 5,758,425.00 *gal* |
| Application event totals | | 81.99 | 9.70 | 99.81 | 668.50 | |

## Field 3 - 05/21/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 80.17 | 9.48 | 97.60 | 611.35 | 528,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 42.74 | 5,691,075.00 *gal* |
| Application event totals | | 80.17 | 9.48 | 97.60 | 654.10 | |

| | | | | |
|---|---|---|---|---|
| 08/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 46.28 | 6,162,525.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 46.28 | |

## Field 4 - 11/19/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough Plant date: 11/19/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 119.04 | 14.08 | 144.92 | 907.77 | 882,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 43.84 | 6,566,625.00 *gal* |
| Application event totals | | 119.04 | 14.08 | 144.92 | 951.61 | |

| | | | | |
|---|---|---|---|---|
| 04/07/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 111.75 | 13.22 | 136.04 | 852.19 | 828,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 40.69 | 6,095,175.00 *gal* |
| Application event totals | | 111.75 | 13.22 | 136.04 | 892.88 | |

## Field 4 - 05/21/2023: Corn, silage

## Field 4 - 05/21/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 05/21/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 44.96 | 6,735,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 44.96 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 97.99 | 11.59 | 119.28 | 747.21 | 726,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 44.29 | 6,633,975.00 *gal* |
| Application event totals | | 97.99 | 11.59 | 119.28 | 791.50 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 44.51 | 6,667,650.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 44.51 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 89.08 | 10.54 | 108.44 | 679.28 | 660,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 47.44 | 7,105,425.00 *gal* |
| Application event totals | | 89.08 | 10.54 | 108.44 | 726.72 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/21/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 79.36 | 9.39 | 96.61 | 605.18 | 588,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 46.99 | 7,038,075.00 *gal* |
| Application event totals | | 79.36 | 9.39 | 96.61 | 652.17 | |

## Field 4 - 05/21/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 79.36 | 9.39 | 96.61 | 605.18 | 588,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 43.84 | 6,566,625.00 *gal* |
| Application event totals | | 79.36 | 9.39 | 96.61 | 649.02 | |

| | | | | |
|---|---|---|---|---|
| 08/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 43.39 | 6,499,275.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 43.39 | |

## Field 5 - 11/19/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough Plant date: 11/19/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/18/2023 | Pipeline | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 131.19 | 15.52 | 159.70 | 1,000.40 | 108,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 50.58 | 841,875.00 *gal* |
| Application event totals | | 131.19 | 15.52 | 159.70 | 1,050.98 | |

| | | | | |
|---|---|---|---|---|
| 04/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 102.03 | 12.07 | 124.21 | 778.09 | 84,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 54.63 | 909,225.00 *gal* |
| Application event totals | | 102.03 | 12.07 | 124.21 | 832.72 | |

## Field 5 - 05/21/2023: Corn, silage

## Field 5 - 05/21/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 05/21/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 52.61 | 875,550.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 52.61 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/07/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 87.46 | 10.35 | 106.47 | 666.93 | 72,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.42 | 606,150.00 *gal* |
| Application event totals | | 87.46 | 10.35 | 106.47 | 703.35 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 52.61 | 875,550.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 52.61 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 87.46 | 10.35 | 106.47 | 666.93 | 72,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 34.40 | 572,475.00 *gal* |
| Application event totals | | 87.46 | 10.35 | 106.47 | 701.33 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 80.17 | 9.48 | 97.60 | 611.35 | 66,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 52.61 | 875,550.00 *gal* |
| Application event totals | | 80.17 | 9.48 | 97.60 | 663.96 | |

## Field 5 - 05/21/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 72.88 | 8.62 | 88.72 | 555.78 | 60,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 50.58 | 841,875.00 *gal* |
| Application event totals | | 72.88 | 8.62 | 88.72 | 606.36 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 38.44 | 639,825.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 38.44 | |

## Field ME - 11/23/2022: Oats, silage-soft dough

Field name: Field ME

Crop: Oats, silage-soft dough Plant date: 11/23/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/06/2022 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Corral solids | 196.81 | 21.22 | 45.34 | 0.00 | 160.00 *ton* |
| Application event totals | | 196.81 | 21.22 | 45.34 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 26.83 | 4.73 | 83.71 | 299.50 | 336,461.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 45.71 | 4,108,350.00 *gal* |
| Application event totals | | 26.83 | 4.73 | 83.71 | 345.21 | |

## Field ME - 11/23/2022: Oats, silage-soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 40.84 | 3,670,575.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 40.84 | |

## Field ME - 06/14/2023: Corn, silage

Field name: Field ME

Crop: Corn, silage

Plant date: 06/14/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.72 | 3,300,150.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 36.72 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/06/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Corral solids | 176.03 | 46.13 | 61.19 | 0.00 | 180.00 *ton* |
| Application event totals | | 176.03 | 46.13 | 61.19 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 84.30 | 4.89 | 136.58 | 840.25 | 448,615.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 37.09 | 3,333,825.00 *gal* |
| Application event totals | | 84.30 | 4.89 | 136.58 | 877.35 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 40.84 | 3,670,575.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 40.84 | |

Field ME - 06/14/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.72 | 3,300,150.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 36.72 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 74.05 | 11.48 | 120.17 | 549.77 | 486,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.34 | 3,266,475.00 *gal* |
| Application event totals | | 74.05 | 11.48 | 120.17 | 586.11 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.97 | 3,232,800.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.97 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 37.09 | 3,333,825.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 37.09 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.34 | 3,266,475.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 36.34 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 37.09 | 3,333,825.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 37.09 | |

## Field MW - 11/23/2022: Oats, silage-soft dough

Field name: Field MW

Crop: Oats, silage-soft dough

Plant date: 11/23/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/06/2022 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Corral solids | 157.32 | 16.97 | 36.24 | 0.00 | 180.00 *ton* |
| Application event totals | | 157.32 | 16.97 | 36.24 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 26.83 | 4.73 | 83.71 | 299.50 | 473,538.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 44.99 | 5,691,075.00 *gal* |
| Application event totals | | 26.83 | 4.73 | 83.71 | 344.49 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 41.53 | 5,253,300.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 41.53 | |

## Field MW - 06/14/2023: Corn, silage

Field name: Field MW

Crop: Corn, silage

Plant date: 06/14/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.41 | 4,478,775.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.41 | |

Field MW - 06/14/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/06/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Corral solids | 173.71 | 45.52 | 60.38 | 0.00 | 250.00 *ton* |
| Application event totals | | 173.71 | 45.52 | 60.38 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 84.30 | 4.89 | 136.58 | 840.25 | 631,384.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.67 | 4,512,450.00 *gal* |
| Application event totals | | 84.30 | 4.89 | 136.58 | 875.93 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.14 | 4,950,225.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 39.14 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.41 | 4,478,775.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.41 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 74.05 | 11.48 | 120.17 | 549.77 | 684,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.14 | 4,445,100.00 *gal* |
| Application event totals | | 74.05 | 11.48 | 120.17 | 584.91 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.14 | 4,950,225.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 39.14 | |

## Field MW - 06/14/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.41 | 4,478,775.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.41 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.41 | 4,478,775.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.41 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.14 | 4,950,225.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 39.14 | |

## Field NE - 11/23/2022: Oats, silage-soft dough

Field name: Field NE

Crop: Oats, silage-soft dough

Plant date: 11/23/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/06/2022 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Corral solids | 181.06 | 47.44 | 62.93 | 0.00 | 240.00 *ton* |
| Application event totals | | 181.06 | 47.44 | 62.93 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 26.85 | 4.73 | 83.77 | 299.70 | 436,453.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 38.44 | 4,478,775.00 *gal* |
| Application event totals | | 26.85 | 4.73 | 83.77 | 338.14 | |

## Field NE - 11/23/2022: Oats, silage-soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 42.49 | 4,950,225.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 42.49 | |

## Field NE - 06/14/2023: Corn, silage

Field name: Field NE

Crop: Corn, silage

Plant date: 06/14/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 38.44 | 4,478,775.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 38.44 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/06/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Corral solids | 184.83 | 48.43 | 64.25 | 0.00 | 245.00 *ton* |
| Application event totals | | 184.83 | 48.43 | 64.25 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 84.30 | 4.89 | 136.58 | 840.25 | 581,538.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.71 | 4,276,725.00 *gal* |
| Application event totals | | 84.30 | 4.89 | 136.58 | 876.96 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 34.97 | 4,074,675.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 34.97 | |

## Field NE - 06/14/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 38.44 | 4,478,775.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 38.44 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 74.05 | 11.48 | 120.17 | 549.77 | 630,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 34.69 | 4,041,000.00 *gal* |
| Application event totals | | 74.05 | 11.48 | 120.17 | 584.45 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.26 | 4,108,350.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.26 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 34.69 | 4,041,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 34.69 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.26 | 4,108,350.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.26 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.55 | 4,142,025.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.55 | |

## Field NW - 11/23/2022: Oats, silage-soft dough

Field name: Field NW

Crop: Oats, silage-soft dough

Plant date: 11/23/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/06/2022 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Corral solids | 170.43 | 18.38 | 39.26 | 0.00 | 195.00 *ton* |
| Application event totals | | 170.43 | 18.38 | 39.26 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 26.83 | 4.73 | 83.71 | 299.50 | 473,538.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.94 | 4,546,125.00 *gal* |
| Application event totals | | 26.83 | 4.73 | 83.71 | 335.44 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.41 | 4,478,775.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.41 | |

## Field NW - 06/14/2023: Corn, silage

Field name: Field NW

Crop: Corn, silage

Plant date: 06/14/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.67 | 4,512,450.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.67 | |

Field NW - 06/14/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/06/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Corral solids | 159.82 | 41.88 | 55.55 | 0.00 | 230.00 *ton* |
| Application event totals | | 159.82 | 41.88 | 55.55 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 84.30 | 4.89 | 136.58 | 840.25 | 631,384.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 32.75 | 4,142,025.00 *gal* |
| Application event totals | | 84.30 | 4.89 | 136.58 | 873.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.41 | 4,478,775.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.41 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 32.21 | 4,074,675.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 32.21 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 74.05 | 11.48 | 120.17 | 549.77 | 684,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.94 | 4,546,125.00 *gal* |
| Application event totals | | 74.05 | 11.48 | 120.17 | 585.71 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 34.61 | 4,377,750.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 34.61 | |

## Field NW - 06/14/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 32.75 | 4,142,025.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 32.75 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.41 | 4,478,775.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.41 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.94 | 4,546,125.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.94 | |

## Field SE - 11/23/2022: Oats, silage-soft dough

Field name: Field SE

Crop: Oats, silage-soft dough

Plant date: 11/23/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/06/2022 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Corral solids | 185.59 | 20.01 | 42.76 | 0.00 | 190.00 *ton* |
| Application event totals | | 185.59 | 20.01 | 42.76 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 26.83 | 4.73 | 83.71 | 299.50 | 423,692.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 43.14 | 4,882,875.00 *gal* |
| Application event totals | | 26.83 | 4.73 | 83.71 | 342.64 | |

## Field SE - 11/23/2022: Oats, silage-soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 43.74 | 4,950,225.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 43.74 | |

## Field SE - 06/14/2023: Corn, silage

Field name: Field SE

Crop: Corn, silage

Plant date: 06/14/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 37.19 | 4,209,375.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 37.19 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/06/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Corral solids | 166.97 | 43.75 | 58.04 | 0.00 | 215.00 *ton* |
| Application event totals | | 166.97 | 43.75 | 58.04 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 84.30 | 4.89 | 136.58 | 840.25 | 564,923.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.00 | 4,074,675.00 *gal* |
| Application event totals | | 84.30 | 4.89 | 136.58 | 876.25 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.00 | 4,074,675.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 36.00 | |

Field SE - 06/14/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.60 | 4,142,025.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 36.60 | |

| | | | | |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 74.05 | 11.48 | 120.17 | 549.77 | 612,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 33.03 | 3,737,925.00 *gal* |
| Application event totals | | 74.05 | 11.48 | 120.17 | 582.80 | |

| | | | | |
|---|---|---|---|---|
| 08/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.00 | 4,074,675.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 36.00 | |

| | | | | |
|---|---|---|---|---|
| 09/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.30 | 4,108,350.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 36.30 | |

| | | | | |
|---|---|---|---|---|
| 09/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 38.38 | 4,344,075.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 38.38 | |

| | | | | |
|---|---|---|---|---|
| 09/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 37.19 | 4,209,375.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 37.19 | |

## Field SW - 11/23/2022: Oats, silage-soft dough

Field name: Field SW

Crop: Oats, silage-soft dough

Plant date: 11/23/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/06/2022 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Corral solids | 166.06 | 17.91 | 38.26 | 0.00 | 180.00 *ton* |
| Application event totals | | 166.06 | 17.91 | 38.26 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 26.83 | 4.73 | 83.71 | 299.50 | 448,615.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 34.57 | 4,142,025.00 *gal* |
| Application event totals | | 26.83 | 4.73 | 83.71 | 334.06 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 34.00 | 4,074,675.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 34.00 | |

## Field SW - 06/14/2023: Corn, silage

Field name: Field SW

Crop: Corn, silage

Plant date: 06/14/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/08/2023 | Sidedress | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.55 | 4,142,025.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.55 | |

Field SW - 06/14/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/06/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Corral solids | 173.51 | 45.47 | 60.31 | 0.00 | 230.00 *ton* |
| Application event totals | | 173.51 | 45.47 | 60.31 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 86.71 | 5.03 | 140.48 | 864.26 | 598,153.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 34.97 | 4,074,675.00 *gal* |
| Application event totals | | 86.71 | 5.03 | 140.48 | 899.23 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.26 | 4,108,350.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.26 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 34.97 | 4,074,675.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 34.97 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| LWI - Hilmar Holsteins | Process wastewater | 76.17 | 11.80 | 123.60 | 565.48 | 648,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.26 | 4,108,350.00 *gal* |
| Application event totals | | 76.17 | 11.80 | 123.60 | 600.74 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.55 | 4,142,025.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.55 | |

Field SW - 06/14/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 34.97 | 4,074,675.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 34.97 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 34.97 | 4,074,675.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 34.97 | |

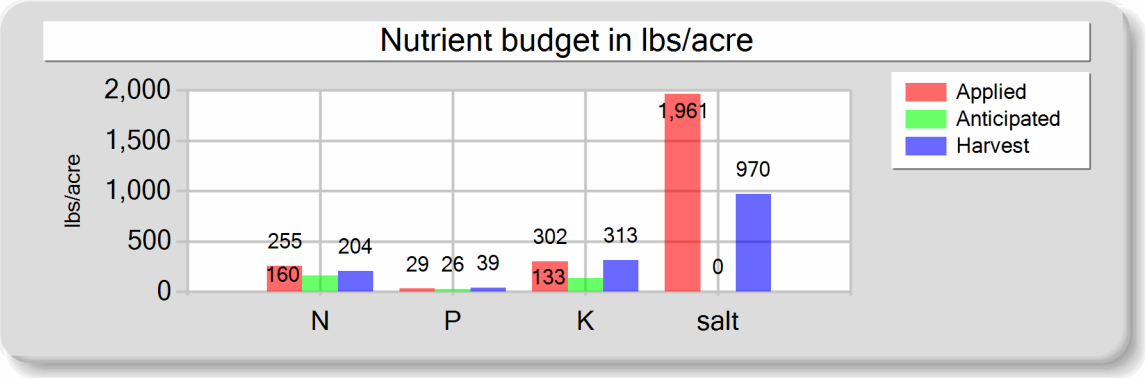
| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.55 | 4,142,025.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.55 | |

## B. NUTRIENT BUDGET

Field 1 - 11/19/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 11/19/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 247.80 | 29.31 | 301.66 | 1,889.64 |
| Fresh water | 0.00 | 0.00 | 0.00 | 71.07 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 254.80 | 29.31 | 301.66 | 1,960.71 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 203.74 | 38.64 | 312.63 | 969.50 |
| Nutrient balance | 51.06 | -9.32 | -10.97 | 991.21 |
| Applied to removed ratio | 1.25 | 0.76 | 0.96 | 2.02 |

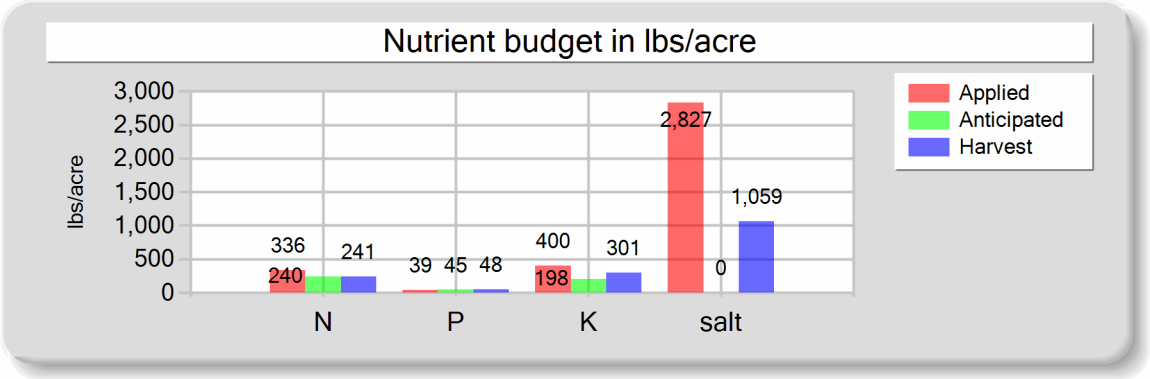
| Fresh water applied | |
|---|---|
| 9,462,675.00 | *gallons* |
| 348.48 | *acre-inches* |
| 8.71 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,632,000.00 | *gallons* |
| 60.10 | *acre-inches* |
| 1.50 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 1 - 05/21/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 05/21/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 328.88 | 38.91 | 400.37 | 2,507.94 |
| Fresh water | 0.00 | 0.00 | 0.00 | 318.67 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 335.88 | 38.91 | 400.37 | 2,826.62 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 241.49 | 48.30 | 300.70 | 1,059.45 |
| Nutrient balance | 94.39 | -9.39 | 99.67 | 1,767.17 |
| Applied to removed ratio | 1.39 | 0.81 | 1.33 | 2.67 |

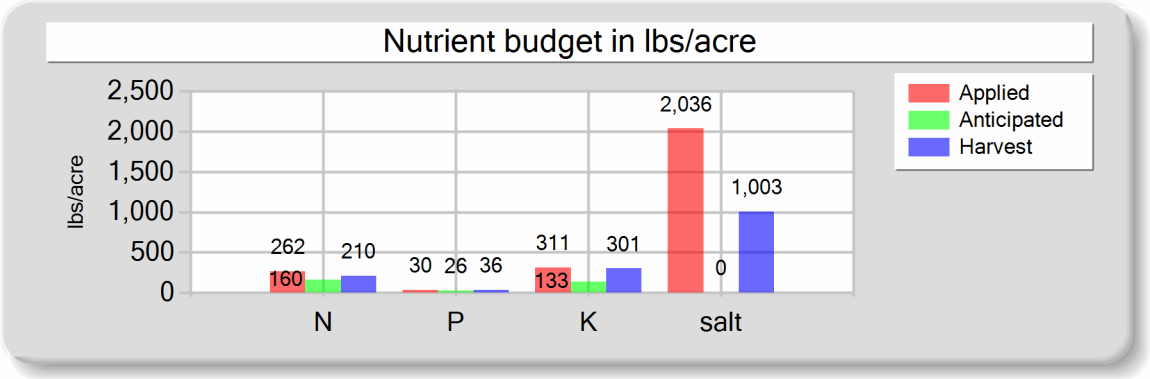
| Fresh water applied | |
|---|---|
| 42,430,500.00 | *gallons* |
| 1,562.57 | *acre-inches* |
| 39.06 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 2,166,000.00 | *gallons* |
| 79.77 | *acre-inches* |
| 1.99 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 2 - 11/19/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 11/19/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 255.09 | 30.18 | 310.53 | 1,945.22 |
| Fresh water | 0.00 | 0.00 | 0.00 | 91.05 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 262.09 | 30.18 | 310.53 | 2,036.27 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 210.27 | 36.21 | 300.78 | 1,002.61 |
| Nutrient balance | 51.82 | -6.03 | 9.75 | 1,033.66 |
| Applied to removed ratio | 1.25 | 0.83 | 1.03 | 2.03 |

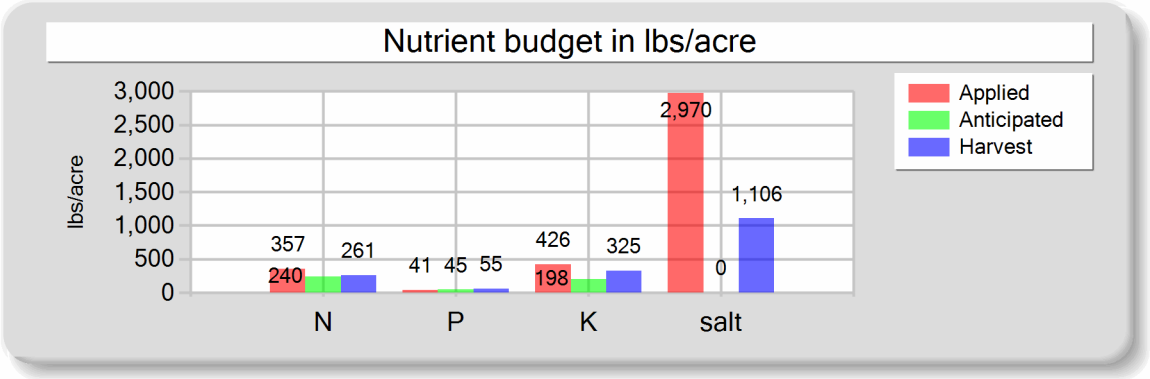
| Fresh water applied | |
|---|---|
| 3,030,750.00 | *gallons* |
| 111.61 | *acre-inches* |
| 11.16 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 420,000.00 | *gallons* |
| 15.47 | *acre-inches* |
| 1.55 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 2 - 05/21/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/21/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 349.83 | 41.39 | 425.88 | 2,667.73 |
| Fresh water | 0.00 | 0.00 | 0.00 | 302.49 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 356.83 | 41.39 | 425.88 | 2,970.22 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 261.04 | 55.32 | 325.00 | 1,106.38 |
| Nutrient balance | 95.80 | -13.93 | 100.88 | 1,863.84 |
| Applied to removed ratio | 1.37 | 0.75 | 1.31 | 2.68 |

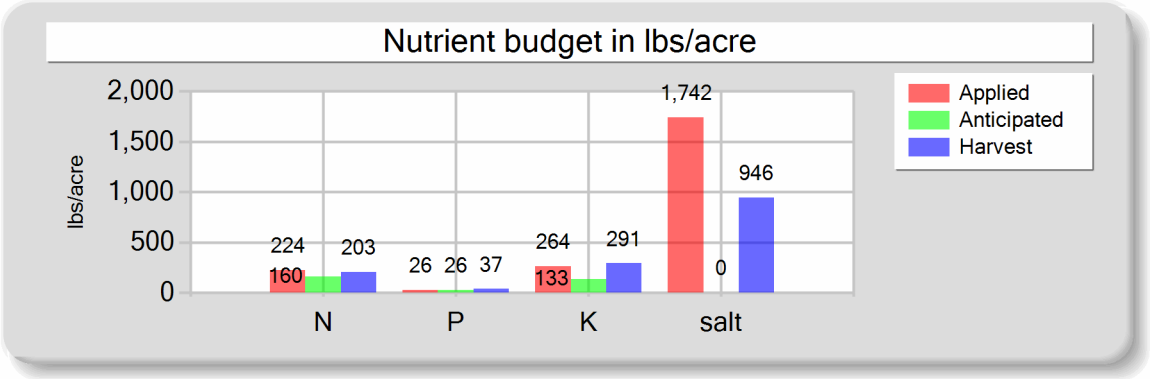
| Fresh water applied | |
|---|---|
| 10,068,825.00 | *gallons* |
| 370.80 | *acre-inches* |
| 37.08 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 576,000.00 | *gallons* |
| 21.21 | *acre-inches* |
| 2.12 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 3 - 11/19/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/19/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 216.82 | 25.65 | 263.95 | 1,653.44 |
| Fresh water | 0.00 | 0.00 | 0.00 | 88.77 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 223.82 | 25.65 | 263.95 | 1,742.21 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 202.58 | 36.65 | 290.54 | 946.25 |
| Nutrient balance | 21.25 | -11.00 | -26.58 | 795.96 |
| Applied to removed ratio | 1.10 | 0.70 | 0.91 | 1.84 |

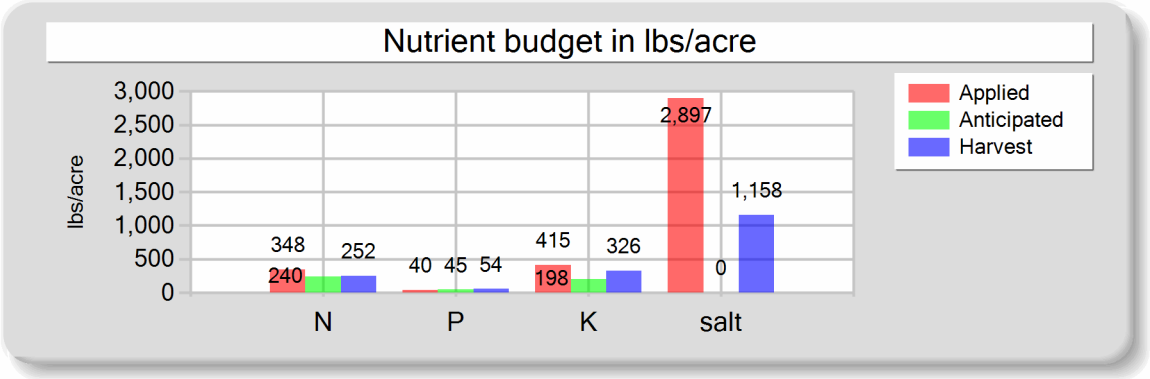
| Fresh water applied | |
|---|---|
| 11,819,925.00 | *gallons* |
| 435.29 | *acre-inches* |
| 10.88 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,428,000.00 | *gallons* |
| 52.59 | *acre-inches* |
| 1.31 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 3 - 05/21/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/21/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 340.72 | 40.31 | 414.78 | 2,598.26 |
| Fresh water | 0.00 | 0.00 | 0.00 | 298.69 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 347.72 | 40.31 | 414.78 | 2,896.95 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 251.68 | 53.69 | 325.50 | 1,157.71 |
| Nutrient balance | 96.05 | -13.38 | 89.28 | 1,739.24 |
| Applied to removed ratio | 1.38 | 0.75 | 1.27 | 2.50 |

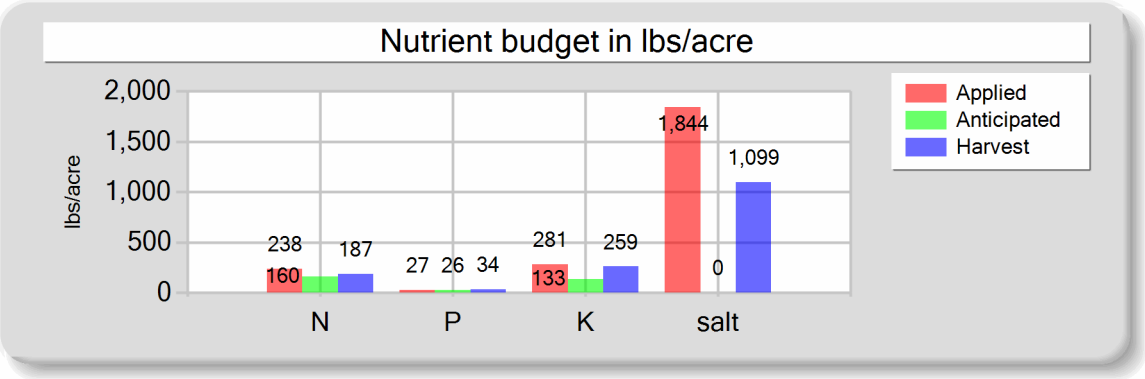
| Fresh water applied | |
|---|---|
| 39,770,175.00 | *gallons* |
| 1,464.60 | *acre-inches* |
| 36.62 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 2,244,000.00 | *gallons* |
| 82.64 | *acre-inches* |
| 2.07 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 4 - 11/19/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 11/19/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 230.79 | 27.30 | 280.96 | 1,759.96 |
| Fresh water | 0.00 | 0.00 | 0.00 | 84.53 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 237.79 | 27.30 | 280.96 | 1,844.49 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 186.86 | 33.59 | 258.92 | 1,099.17 |
| Nutrient balance | 50.93 | -6.28 | 22.04 | 745.32 |
| Applied to removed ratio | 1.27 | 0.81 | 1.09 | 1.68 |

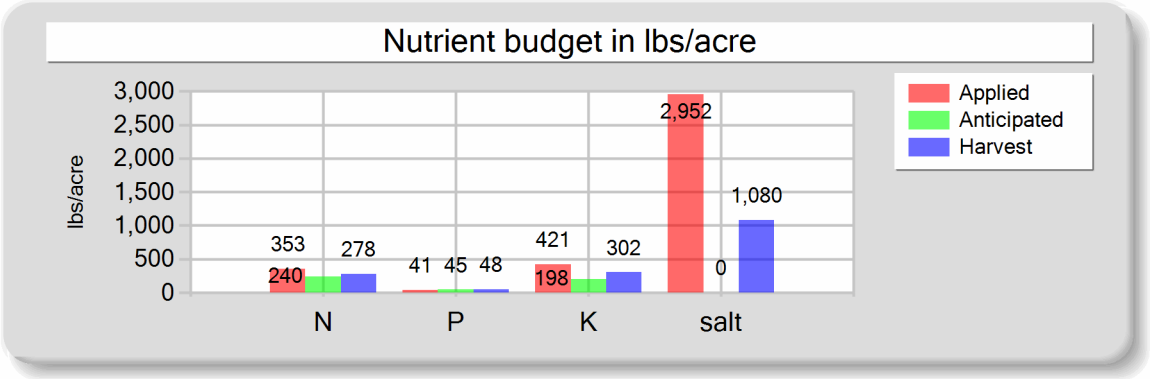
| Fresh water applied | |
|---|---|
| 12,661,800.00 | *gallons* |
| 466.29 | *acre-inches* |
| 10.36 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,710,000.00 | *gallons* |
| 62.97 | *acre-inches* |
| 1.40 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 4 - 05/21/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 05/21/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 345.78 | 40.91 | 420.95 | 2,636.85 |
| Fresh water | 0.00 | 0.00 | 0.00 | 315.41 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 352.78 | 40.91 | 420.95 | 2,952.27 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 277.77 | 47.84 | 302.46 | 1,080.20 |
| Nutrient balance | 75.02 | -6.93 | 118.49 | 1,872.06 |
| Applied to removed ratio | 1.27 | 0.86 | 1.39 | 2.73 |

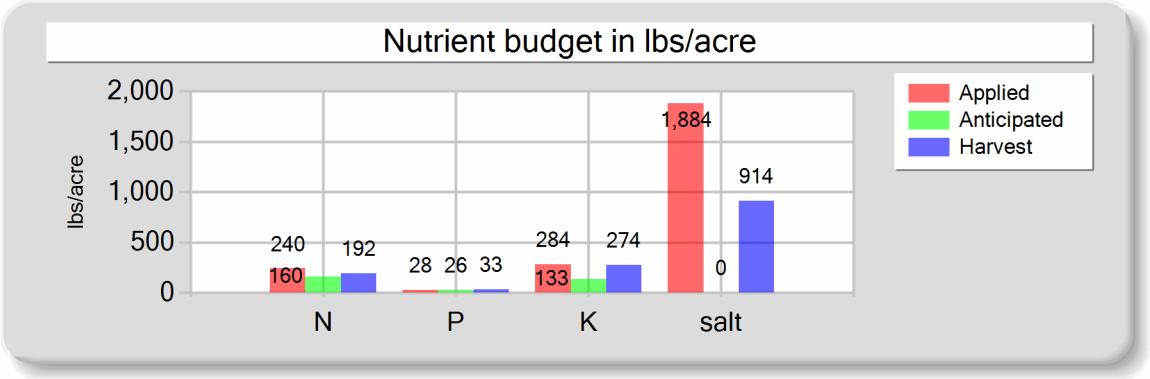
| Fresh water applied | |
|---|---|
| 47,246,025.00 | *gallons* |
| 1,739.91 | *acre-inches* |
| 38.66 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 2,562,000.00 | *gallons* |
| 94.35 | *acre-inches* |
| 2.10 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 5 - 11/19/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 11/19/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 233.22 | 27.59 | 283.92 | 1,778.49 |
| Fresh water | 0.00 | 0.00 | 0.00 | 105.21 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 240.22 | 27.59 | 283.92 | 1,883.70 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 191.77 | 33.02 | 274.32 | 914.41 |
| Nutrient balance | 48.45 | -5.43 | 9.59 | 969.29 |
| Applied to removed ratio | 1.25 | 0.84 | 1.03 | 2.06 |

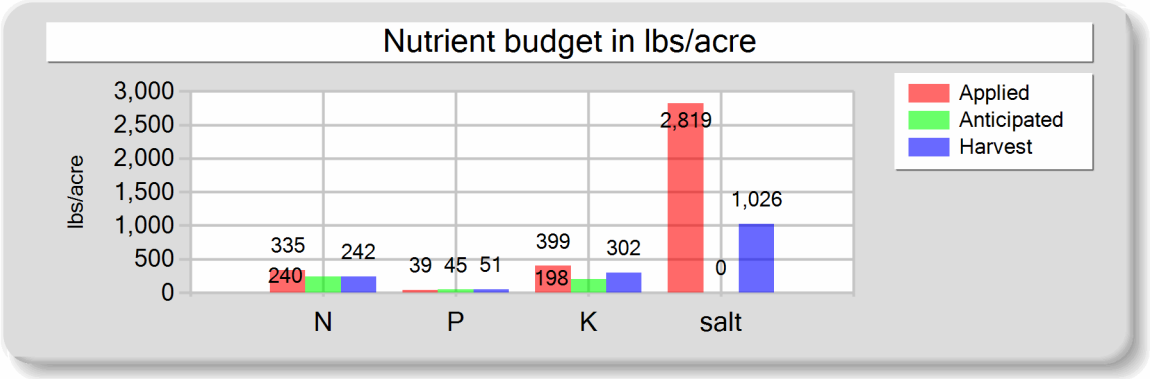
| Fresh water applied | |
|---|---|
| 1,751,100.00 | *gallons* |
| 64.49 | *acre-inches* |
| 12.90 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 192,000.00 | *gallons* |
| 7.07 | *acre-inches* |
| 1.41 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 5 - 05/21/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 05/21/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 327.97 | 38.80 | 399.26 | 2,501.00 |
| Fresh water | 0.00 | 0.00 | 0.00 | 317.66 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 334.97 | 38.80 | 399.26 | 2,818.66 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 242.18 | 51.32 | 301.52 | 1,026.46 |
| Nutrient balance | 92.79 | -12.52 | 97.74 | 1,792.20 |
| Applied to removed ratio | 1.38 | 0.76 | 1.32 | 2.75 |

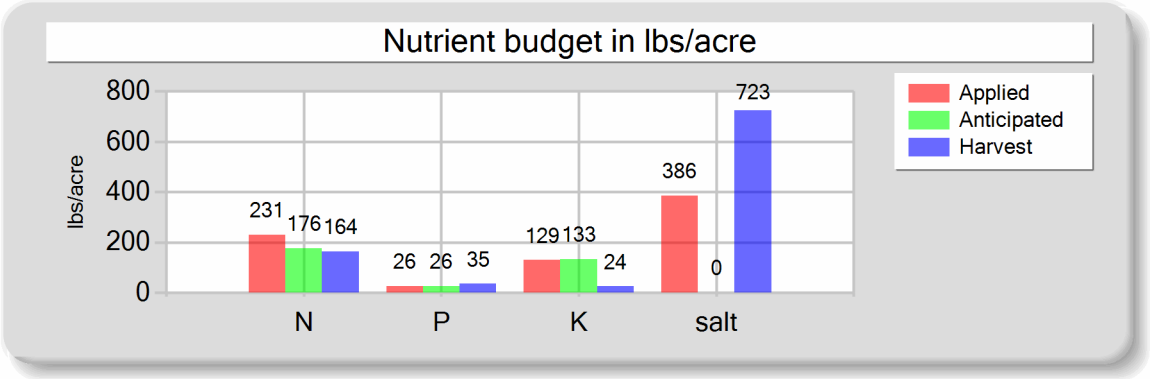
| Fresh water applied | |
|---|---|
| 5,286,975.00 | *gallons* |
| 194.70 | *acre-inches* |
| 38.94 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 270,000.00 | *gallons* |
| 9.94 | *acre-inches* |
| 1.99 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field ME - 11/23/2022: Oats, silage-soft dough

Field name: Field ME  Crop: Oats, silage-soft dough  Plant date: 11/23/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 196.81 | 21.22 | 45.34 | 0.00 |
| Process wastewater | 26.83 | 4.73 | 83.71 | 299.50 |
| Fresh water | 0.00 | 0.00 | 0.00 | 86.55 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 230.64 | 25.96 | 129.06 | 386.05 |
| Anticipated crop nutrient removal | 176.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 164.41 | 34.76 | 24.43 | 723.41 |
| Nutrient balance | 66.23 | -8.81 | 104.63 | -337.36 |
| Applied to removed ratio | 1.40 | 0.75 | 5.28 | 0.53 |

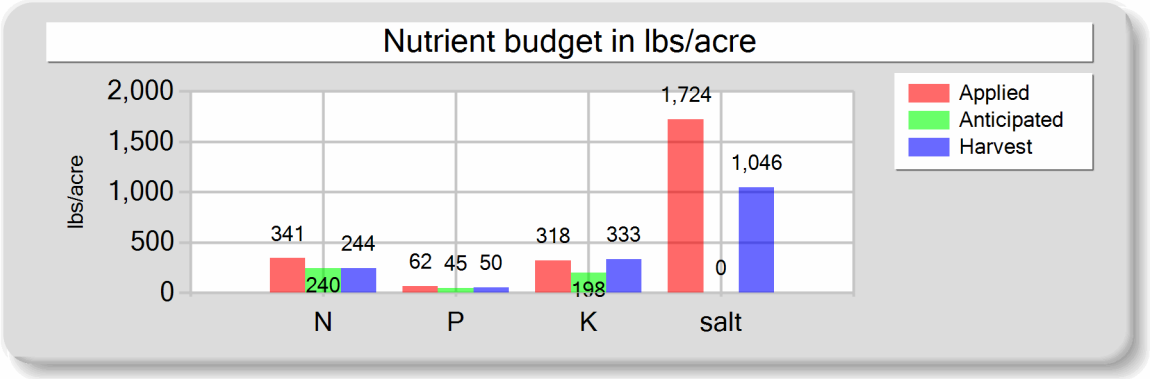
| Fresh water applied | |
|---|---|
| 7,778,925.00 | *gallons* |
| 286.47 | *acre-inches* |
| 10.61 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 336,461.00 | *gallons* |
| 12.39 | *acre-inches* |
| 0.46 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field ME - 06/14/2023: Corn, silage

Field name: Field ME  Crop: Corn, silage  Plant date: 06/14/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 176.03 | 46.13 | 61.19 | 0.00 |
| Process wastewater | 158.36 | 16.37 | 256.74 | 1,390.02 |
| Fresh water | 0.00 | 0.00 | 0.00 | 334.22 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 341.39 | 62.50 | 317.93 | 1,724.24 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 244.13 | 50.13 | 333.50 | 1,046.27 |
| Nutrient balance | 97.26 | 12.36 | -15.57 | 677.98 |
| Applied to removed ratio | 1.40 | 1.25 | 0.95 | 1.65 |

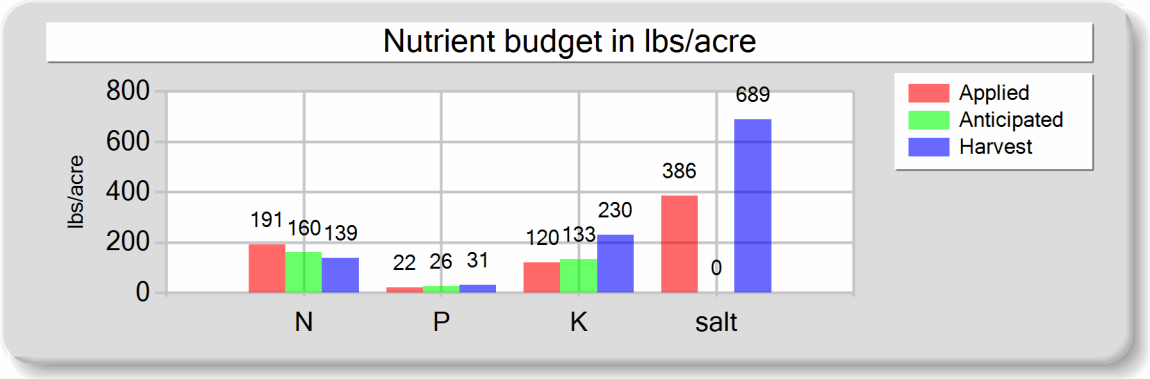
| Fresh water applied | |
|---|---|
| 30,038,100.00 | *gallons* |
| 1,106.20 | *acre-inches* |
| 40.97 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 934,615.00 | *gallons* |
| 34.42 | *acre-inches* |
| 1.27 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field MW - 11/23/2022: Oats, silage-soft dough

Field name: Field MW  Crop: Oats, silage-soft dough  Plant date: 11/23/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 157.32 | 16.97 | 36.24 | 0.00 |
| Process wastewater | 26.83 | 4.73 | 83.71 | 299.50 |
| Fresh water | 0.00 | 0.00 | 0.00 | 86.52 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 191.15 | 21.70 | 119.96 | 386.02 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 138.67 | 30.71 | 229.87 | 688.68 |
| Nutrient balance | 52.48 | -9.01 | -109.91 | -302.66 |
| Applied to removed ratio | 1.38 | 0.71 | 0.52 | 0.56 |

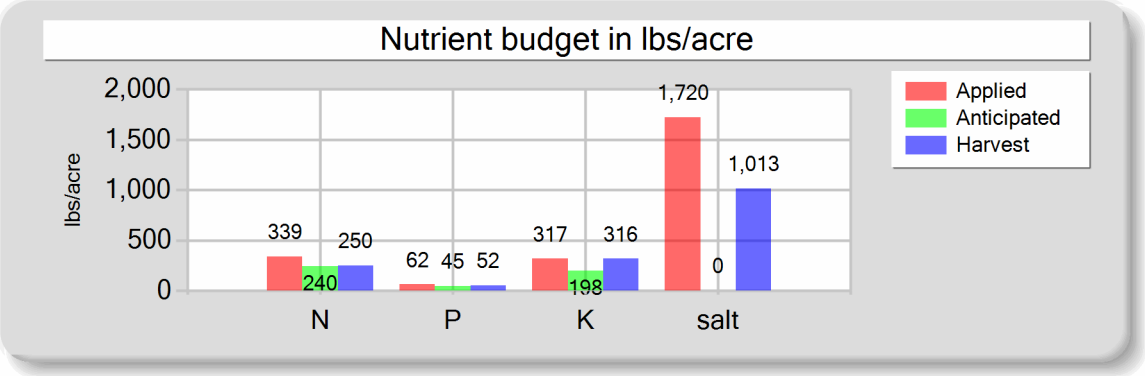
| Fresh water applied | |
|---|---|
| 10,944,375.00 | *gallons* |
| 403.04 | *acre-inches* |
| 10.61 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 473,538.00 | *gallons* |
| 17.44 | *acre-inches* |
| 0.46 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field MW - 06/14/2023: Corn, silage

Field name: Field MW  Crop: Corn, silage  Plant date: 06/14/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 173.71 | 45.52 | 60.38 | 0.00 |
| Process wastewater | 158.36 | 16.37 | 256.74 | 1,390.02 |
| Fresh water | 0.00 | 0.00 | 0.00 | 329.86 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 339.07 | 61.89 | 317.12 | 1,719.88 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 250.22 | 51.70 | 316.39 | 1,013.27 |
| Nutrient balance | 88.85 | 10.19 | 0.74 | 706.60 |
| Applied to removed ratio | 1.36 | 1.20 | 1.00 | 1.70 |

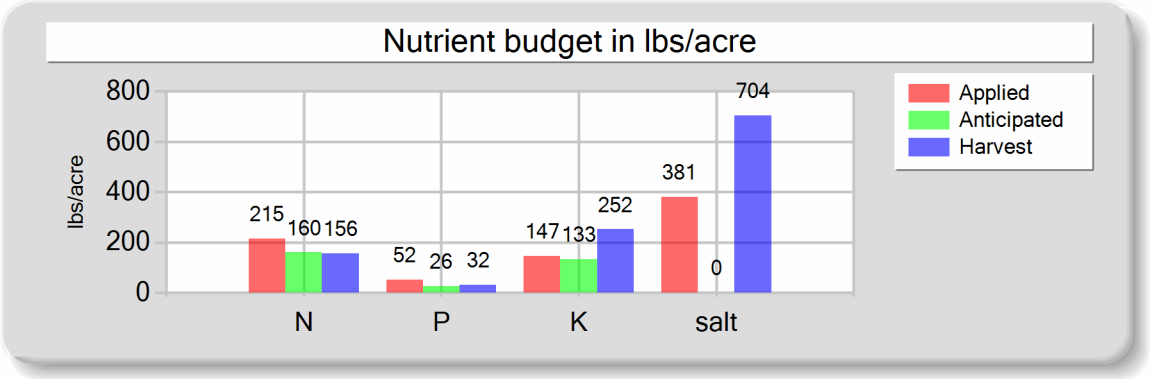
| Fresh water applied | |
|---|---|
| 41,723,325.00 | *gallons* |
| 1,536.53 | *acre-inches* |
| 40.43 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,315,384.00 | *gallons* |
| 48.44 | *acre-inches* |
| 1.27 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field NE - 11/23/2022: Oats, silage-soft dough

Field name: Field NE  Crop: Oats, silage-soft dough  Plant date: 11/23/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 181.06 | 47.44 | 62.93 | 0.00 |
| Process wastewater | 26.85 | 4.73 | 83.77 | 299.70 |
| Fresh water | 0.00 | 0.00 | 0.00 | 80.93 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 214.91 | 52.18 | 146.71 | 380.63 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 156.09 | 31.58 | 251.74 | 703.78 |
| Nutrient balance | 58.81 | 20.60 | -105.03 | -323.15 |
| Applied to removed ratio | 1.38 | 1.65 | 0.58 | 0.54 |

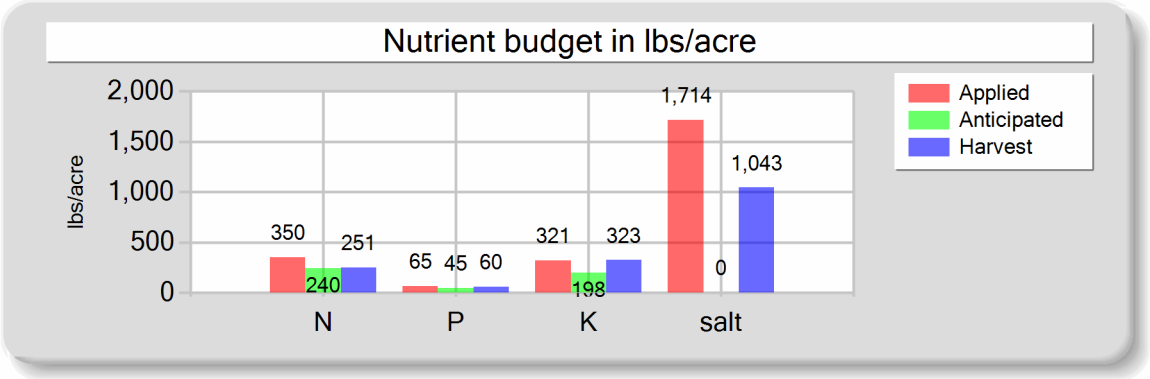
| Fresh water applied | |
|---|---|
| 9,429,000.00 | *gallons* |
| 347.24 | *acre-inches* |
| 9.92 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 436,453.00 | *gallons* |
| 16.07 | *acre-inches* |
| 0.46 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field NE - 06/14/2023: Corn, silage

Field name: Field NE  Crop: Corn, silage  Plant date: 06/14/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 184.83 | 48.43 | 64.25 | 0.00 |
| Process wastewater | 158.36 | 16.37 | 256.74 | 1,390.02 |
| Fresh water | 0.00 | 0.00 | 0.00 | 324.02 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 350.19 | 64.80 | 320.99 | 1,714.04 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 251.06 | 59.57 | 323.40 | 1,042.55 |
| Nutrient balance | 99.12 | 5.23 | -2.42 | 671.49 |
| Applied to removed ratio | 1.39 | 1.09 | 0.99 | 1.64 |

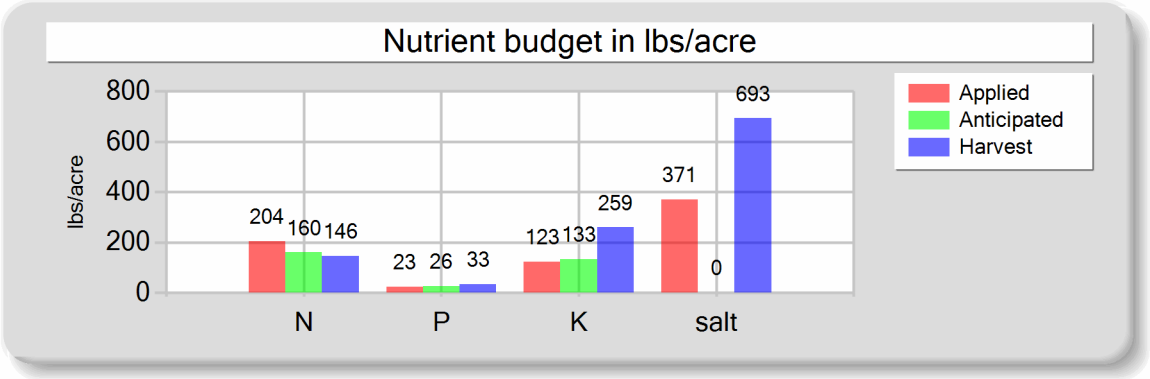
| Fresh water applied | |
|---|---|
| 37,749,675.00 | *gallons* |
| 1,390.19 | *acre-inches* |
| 39.72 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,211,538.00 | *gallons* |
| 44.62 | *acre-inches* |
| 1.27 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field NW - 11/23/2022: Oats, silage-soft dough

Field name: Field NW  Crop: Oats, silage-soft dough  Plant date: 11/23/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 170.43 | 18.38 | 39.26 | 0.00 |
| Process wastewater | 26.83 | 4.73 | 83.71 | 299.50 |
| Fresh water | 0.00 | 0.00 | 0.00 | 71.35 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 204.26 | 23.11 | 122.98 | 370.84 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 146.17 | 33.22 | 259.12 | 692.88 |
| Nutrient balance | 58.09 | -10.11 | -136.14 | -322.03 |
| Applied to removed ratio | 1.40 | 0.70 | 0.47 | 0.54 |

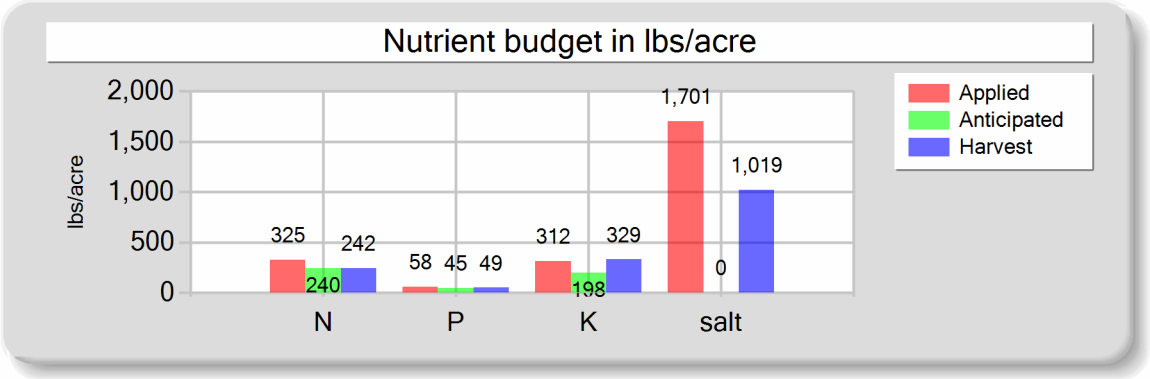
| Fresh water applied | |
|---|---|
| 9,024,900.00 | *gallons* |
| 332.36 | *acre-inches* |
| 8.75 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 473,538.00 | *gallons* |
| 17.44 | *acre-inches* |
| 0.46 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field NW - 06/14/2023: Corn, silage

Field name: Field NW  Crop: Corn, silage  Plant date: 06/14/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 159.82 | 41.88 | 55.55 | 0.00 |
| Process wastewater | 158.36 | 16.37 | 256.74 | 1,390.02 |
| Fresh water | 0.00 | 0.00 | 0.00 | 310.69 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 325.17 | 58.25 | 312.29 | 1,700.71 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 241.98 | 48.82 | 329.01 | 1,018.86 |
| Nutrient balance | 83.19 | 9.43 | -16.71 | 681.85 |
| Applied to removed ratio | 1.34 | 1.19 | 0.95 | 1.67 |

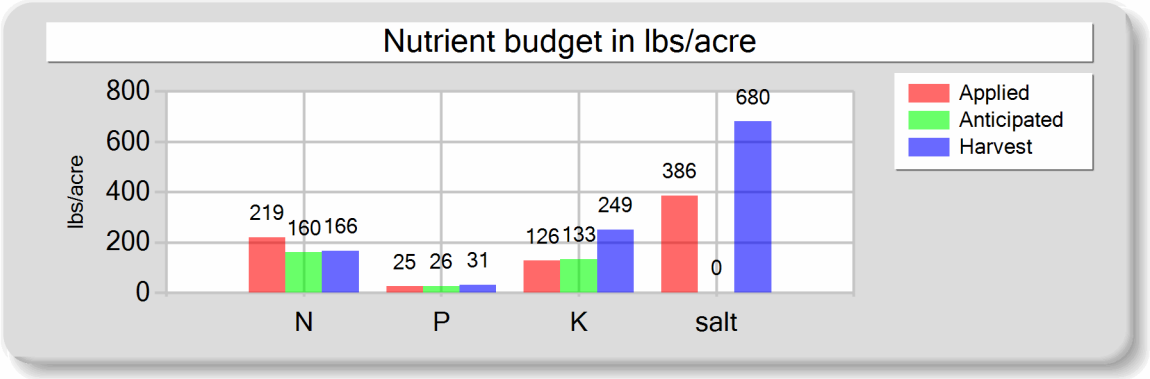
| Fresh water applied | |
|---|---|
| 39,298,725.00 | *gallons* |
| 1,447.24 | *acre-inches* |
| 38.09 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,315,384.00 | *gallons* |
| 48.44 | *acre-inches* |
| 1.27 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field SE - 11/23/2022: Oats, silage-soft dough

Field name: Field SE  Crop: Oats, silage-soft dough  Plant date: 11/23/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 185.59 | 20.01 | 42.76 | 0.00 |
| Process wastewater | 26.83 | 4.73 | 83.71 | 299.50 |
| Fresh water | 0.00 | 0.00 | 0.00 | 86.88 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 219.42 | 24.75 | 126.47 | 386.38 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 165.68 | 31.26 | 249.30 | 679.90 |
| Nutrient balance | 53.75 | -6.51 | -122.82 | -293.52 |
| Applied to removed ratio | 1.32 | 0.79 | 0.51 | 0.57 |

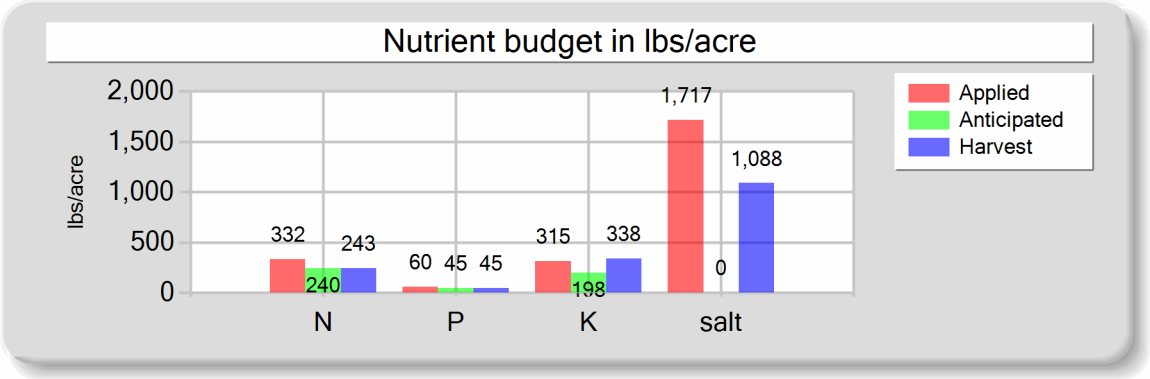
| Fresh water applied | |
|---|---|
| 9,833,100.00 | *gallons* |
| 362.12 | *acre-inches* |
| 10.65 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 423,692.00 | *gallons* |
| 15.60 | *acre-inches* |
| 0.46 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field SE - 06/14/2023: Corn, silage

Field name: Field SE  Crop: Corn, silage  Plant date: 06/14/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 166.97 | 43.75 | 58.04 | 0.00 |
| Process wastewater | 158.36 | 16.37 | 256.74 | 1,390.02 |
| Fresh water | 0.00 | 0.00 | 0.00 | 326.71 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 332.32 | 60.12 | 314.78 | 1,716.73 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 242.63 | 45.35 | 337.87 | 1,088.45 |
| Nutrient balance | 89.69 | 14.77 | -23.09 | 628.28 |
| Applied to removed ratio | 1.37 | 1.33 | 0.93 | 1.58 |

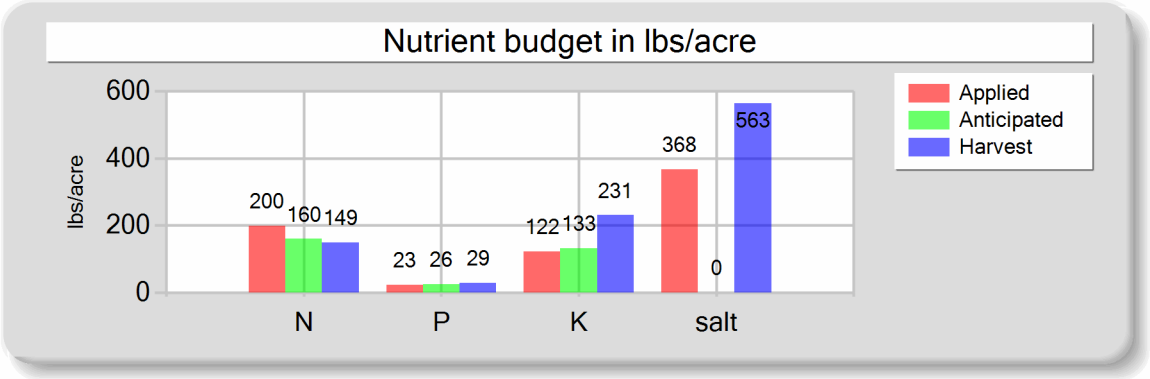
| Fresh water applied | |
|---|---|
| 36,975,150.00 | *gallons* |
| 1,361.67 | *acre-inches* |
| 40.05 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,176,923.00 | *gallons* |
| 43.34 | *acre-inches* |
| 1.27 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field SW - 11/23/2022: Oats, silage-soft dough

Field name: Field SW  Crop: Oats, silage-soft dough  Plant date: 11/23/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 166.06 | 17.91 | 38.26 | 0.00 |
| Process wastewater | 26.83 | 4.73 | 83.71 | 299.50 |
| Fresh water | 0.00 | 0.00 | 0.00 | 68.57 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 199.89 | 22.64 | 121.97 | 368.06 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 148.76 | 28.55 | 230.65 | 563.48 |
| Nutrient balance | 51.13 | -5.91 | -108.68 | -195.41 |
| Applied to removed ratio | 1.34 | 0.79 | 0.53 | 0.65 |

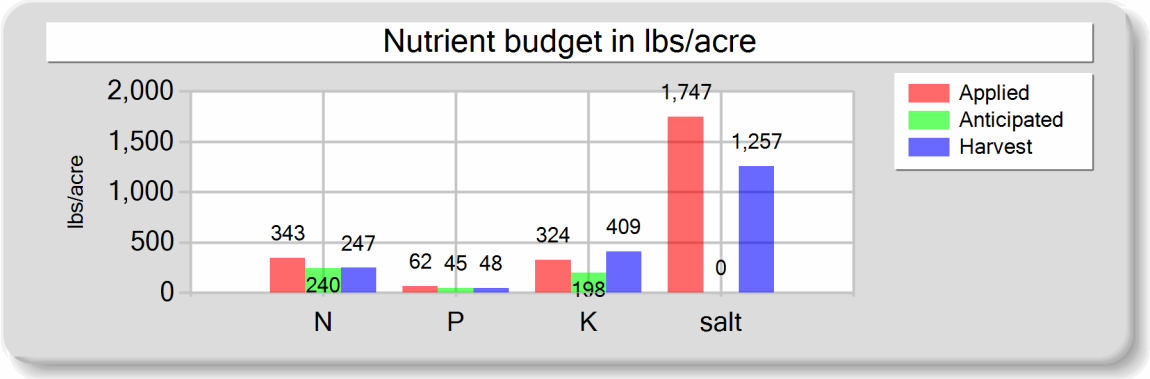
| Fresh water applied | |
|---|---|
| 8,216,700.00 | *gallons* |
| 302.59 | *acre-inches* |
| 8.41 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 448,615.00 | *gallons* |
| 16.52 | *acre-inches* |
| 0.46 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field SW - 06/14/2023: Corn, silage

Field name: Field SW  Crop: Corn, silage  Plant date: 06/14/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 173.51 | 45.47 | 60.31 | 0.00 |
| Process wastewater | 162.88 | 16.84 | 264.08 | 1,429.73 |
| Fresh water | 0.00 | 0.00 | 0.00 | 317.08 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 343.39 | 62.30 | 324.39 | 1,746.82 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 246.82 | 47.99 | 409.08 | 1,256.95 |
| Nutrient balance | 96.58 | 14.31 | -84.69 | 489.87 |
| Applied to removed ratio | 1.39 | 1.30 | 0.79 | 1.39 |

| Fresh water applied | |
|---|---|
| 36,941,475.00 | *gallons* |
| 1,360.43 | *acre-inches* |
| 38.87 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,246,153.00 | *gallons* |
| 45.89 | *acre-inches* |
| 1.31 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

NUTRIENT ANALYSES

## A. MANURE ANALYSES

### Solid Manure

Sample and source description: Solid Manure
Sample date: 05/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 18.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 20,400.00 | 2,200.00 | 4,700.00 | | | | | | | |
| **DL** | 100.00 | 100.00 | 100.00 | | | | | | | |

### Solid Manure

Sample and source description: Solid Manure
Sample date: 11/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 29.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 18,700.00 | 4,900.00 | 6,500.00 | | | | | | | |
| **DL** | 100.00 | 100.00 | 100.00 | | | | | | | |

## B. PROCESS WASTEWATER ANALYSES

### Lagoon Water

Sample and source description: Lagoon Water
Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 726.00 | 468.00 | 0.00 | 1.80 | 86.10 | 886.00 | | | | | | | | 11,100.00 | 5,550 |
| **DL** | 1.00 | 0.50 | 0.50 | 0.10 | 0.10 | 0.50 | | | | | | | | 10.00 | 10 |

## Lagoon Water - Import From Hilmar Holsteins

Sample and source description: Lagoon Water - Import From Hilmar Holsteins

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 258.00 | 0.00 | 0.00 | 0.00 | 45.50 | 805.00 | | | | | | | | 5,060.00 | 2,880 |
| **DL** | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | | | | | | | | 1.00 | 1 |

## Lagoon Water - Import From Hilmar Holsteins

Sample and source description: Lagoon Water - Import From Hilmar Holsteins

Sample date: 06/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 608.00 | 0.00 | 0.00 | 0.00 | 35.30 | 985.00 | | | | | | | | 7,730.00 | 6,060 |
| **DL** | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | | | | | | | | 1.00 | 1 |

## Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 789.00 | 481.00 | 0.00 | 1.10 | 47.90 | 450.00 | | | | | | | | 12,300.00 | 5,750 |
| **DL** | 1.00 | 1.00 | 1.00 | 1.00 | 0.10 | 0.50 | | | | | | | | 10.00 | 10 |

## Lagoon Water - Import From Hilmar Holsteins

Sample and source description: Lagoon Water - Import From Hilmar Holsteins

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 493.00 | 0.00 | 0.00 | 0.00 | 76.40 | 800.00 | | | | | | | | 6,910.00 | 3,660 |
| **DL** | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | | | | | | | | 1.00 | 1 |

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 1,050.00 | 434.00 | 0.00 | 0.90 | 183.00 | 817.00 | | | | | | | | 10,800.00 | 5,250 |
| **DL** | 1.00 | 0.50 | 0.50 | 0.10 | 0.10 | 0.50 | | | | | | | | 10.00 | 10 |

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 826.00 | 491.00 | 0.00 | 0.10 | 185.00 | 1,010.00 | | | | | | | | 12,000.00 | 6,650 |
| **DL** | 1.00 | 0.50 | 1.00 | 0.10 | 0.10 | 0.50 | | | | | | | | 10.00 | 10 |

## C. FRESH WATER ANALYSES

Domestic Well - Domestic Well East

Domestic Well East

Sample description: Domestic Well East

Sample date: 11/15/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 53.10 | 0.00 | 53.10 | | | | | | | | 1,640.00 | 0 |
| **DL** | 0.10 | 1.00 | 0.10 | | | | | | | | 10.00 | 1 |

Domestic Well - Domestic Well West

## Domestic Well - Domestic Well West

### Domestic Well West

Sample description: Domestic Well West

Sample date: 11/15/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 62.70 | 0.00 | 62.70 | | | | | | | | 1,530.00 | 0 |
| **DL** | 0.10 | 1.00 | 0.10 | | | | | | | | 10.00 | 1 |

## Irrigation Pump

### Irrigation Pump

Sample description: Irrigation Pump

Sample date: 08/21/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 23.90 | 0.00 | 23.90 | | | | | | | | 711.00 | 545 |
| **DL** | 1.00 | 1.00 | 0.10 | | | | | | | | 10.00 | 10 |

## Irrigation Water - Canal

### Irrigation water - Canal Water

Sample description: Irrigation water - Canal Water

Sample date: 08/02/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 0.00 | 0.00 | 0.00 | | | | | | | | 31.80 | 36 |
| **DL** | 0.01 | 0.01 | 0.01 | | | | | | | | 10.00 | 10 |

**D. SOIL ANALYSES**

## Field 1

### 1

Sample and source description: 1

Sample date: 11/07/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 229.00 | | | | |
| **DL** | | | 2.00 | | | | |

## Field 2

### 2

Sample and source description: 2

Sample date: 11/21/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 126.00 | | | | |
| **DL** | | | 2.00 | | | | |

## Field 3

### 3

Sample and source description: 3

Sample date: 11/21/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 155.00 | | | | |
| **DL** | | | 2.00 | | | | |

## Field 4

## Field 4

### 4

Sample and source description: 4

Sample date: 11/21/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 137.00 | | | | |
| **DL** | | | 2.00 | | | | |

## Field 5

### 5

Sample and source description: 5

Sample date: 11/21/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 170.00 | | | | |
| **DL** | | | 2.00 | | | | |

## Field ME

### ME

Sample and source description: ME

Sample date: 11/21/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 141.00 | | | | |
| **DL** | | | 2.00 | | | | |

## Field MW

## Field MW

### ME

Sample and source description: ME

Sample date: 11/21/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 105.00 | | | | |
| **DL** | | | 2.00 | | | | |

## Field NE

### NE

Sample and source description: NE

Sample date: 11/13/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 220.00 | | | | |
| **DL** | | | 2.00 | | | | |

## Field NW

### NW

Sample and source description: NW

Sample date: 11/07/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 149.00 | | | | |
| **DL** | | | 2.00 | | | | |

## Field SE

Field SE

SE

Sample and source description: SE

Sample date: 11/30/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 129.00 | | | | |
| **DL** | | | 2.00 | | | | |

Field SW

SW

Sample and source description: SW

Sample date: 11/30/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 122.00 | | | | |
| **DL** | | | 2.00 | | | | |

## E. PLANT TISSUE ANALYSES

Field 1 - 11/19/2022: Oats, silage-soft dough

1

Sample and source description: 1

Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.1 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 29,000.00 | 5,500.00 | 44,500.00 | | 13.80 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

Field 1 - 05/21/2023: Corn, silage

## Field 1 - 05/21/2023: Corn, silage

### Field 1

Sample and source description: Field 1

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 15,500.00 | 3,100.00 | 19,300.00 | | 6.80 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 2 - 11/19/2022: Oats, silage-soft dough

### 2

Sample and source description: 2

Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 30,200.00 | 5,200.00 | 43,200.00 | | 14.40 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 2 - 05/21/2023: Corn, silage

### Field 2 & 5

Sample and source description: Field 2 & 5

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 15,100.00 | 3,200.00 | 18,800.00 | | 6.40 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 3 - 11/19/2022: Oats, silage-soft dough

### 3

Sample and source description: 3

Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 30,400.00 | 5,500.00 | 43,600.00 | | 14.20 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 3 - 05/21/2023: Corn, silage

### Field 3

Sample and source description: Field 3

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.1 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 15,000.00 | 3,200.00 | 19,400.00 | | 6.90 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 4 - 11/19/2022: Oats, silage-soft dough

### 4A

Sample and source description: 4A

Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.1 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 30,600.00 | 5,500.00 | 42,400.00 | | 18.00 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 4 - 11/19/2022: Oats, silage-soft dough

### 4B

Sample and source description: 4B

Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 29,700.00 | 4,600.00 | 37,000.00 | | 15.70 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 4 - 05/21/2023: Corn, silage

### Field 4 A

Sample and source description: Field 4 A

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 18,000.00 | 3,100.00 | 19,600.00 | | 7.00 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

### Field 4 B

Sample and source description: Field 4 B

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 15,000.00 | 3,300.00 | 19,900.00 | | 6.70 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 5 - 11/19/2022: Oats, silage-soft dough

Field 5 - 11/19/2022: Oats, silage-soft dough

5

Sample and source description: 5

Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 30,200.00 | 5,200.00 | 43,200.00 | | 14.40 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

Field 5 - 05/21/2023: Corn, silage

Field 2 & 5

Sample and source description: Field 2 & 5

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 15,100.00 | 3,200.00 | 18,800.00 | | 6.40 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

Field ME - 11/23/2022: Oats, silage-soft dough

Field ME

Sample and source description: Field ME

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.5 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 17,500.00 | 3,700.00 | 2,600.00 | | 7.70 |
| **DL** | 0.01 | 0.01 | 0.01 | | 0.01 |

## Field ME - 06/14/2023: Corn, silage

### Field ME

Sample and source description: Field ME

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.7 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 11,200.00 | 2,300.00 | 15,300.00 | | 4.80 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field MW - 11/23/2022: Oats, silage-soft dough

### Field MW

Sample and source description: Field MW

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 14,900.00 | 3,300.00 | 24,700.00 | | 7.40 |
| **DL** | 0.01 | 0.01 | 0.01 | | 0.01 |

## Field MW - 06/14/2023: Corn, silage

### Field MW

Sample and source description: Field MW

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 12,100.00 | 2,500.00 | 15,300.00 | | 4.90 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field NE - 11/23/2022: Oats, silage-soft dough

### Field NE

Sample and source description: Field NE

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 17,300.00 | 3,500.00 | 27,900.00 | | 7.80 |
| **DL** | 0.01 | 0.01 | 0.01 | | 0.01 |

## Field NE - 06/14/2023: Corn, silage

### Field NE

Sample and source description: Field NE

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 11,800.00 | 2,800.00 | 15,200.00 | | 4.90 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

### Field NW

Sample and source description: Field NW

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 11,400.00 | 2,300.00 | 15,500.00 | | 4.80 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field NW - 11/23/2022: Oats, silage-soft dough

## Field NW - 11/23/2022: Oats, silage-soft dough

### Field NW

Sample and source description: Field NW

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 15,400.00 | 3,500.00 | 27,300.00 | | 7.30 |
| **DL** | 0.01 | 0.01 | 0.01 | | 0.01 |

## Field NW - 06/14/2023: Corn, silage

### Field NW

Sample and source description: Field NW

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 11,400.00 | 2,300.00 | 15,500.00 | | 4.80 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field SE - 11/23/2022: Oats, silage-soft dough

### Field SE

Sample and source description: Field SE

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.9 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 21,200.00 | 4,000.00 | 31,900.00 | | 8.70 |
| **DL** | 0.01 | 0.01 | 0.01 | | 0.01 |

## Field SE - 06/14/2023: Corn, silage

### Field SE

Sample and source description: Field SE

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 10,700.00 | 2,000.00 | 14,900.00 | | 4.80 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field SW - 11/23/2022: Oats, silage-soft dough

### Field SW

Sample and source description: Field SW

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 19,800.00 | 3,800.00 | 30,700.00 | | 7.50 |
| **DL** | 0.01 | 0.01 | 0.01 | | 0.01 |

## Field SW - 06/14/2023: Corn, silage

### Field SW

Sample and source description: Field SW

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.5 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 10,800.00 | 2,100.00 | 17,900.00 | | 5.50 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## F. SUBSURFACE (TILE) DRAINAGE ANALYSES

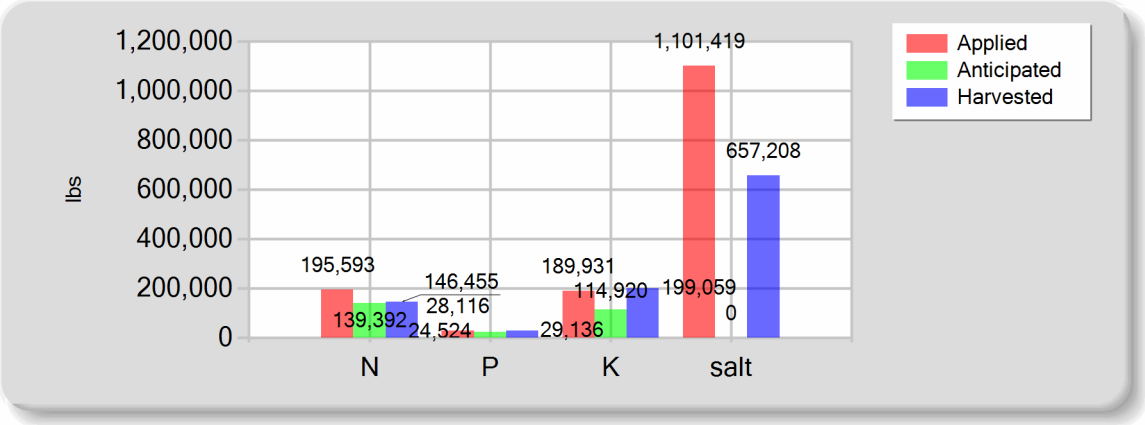
*No subsurface (tile) drainage analyses entered.*

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

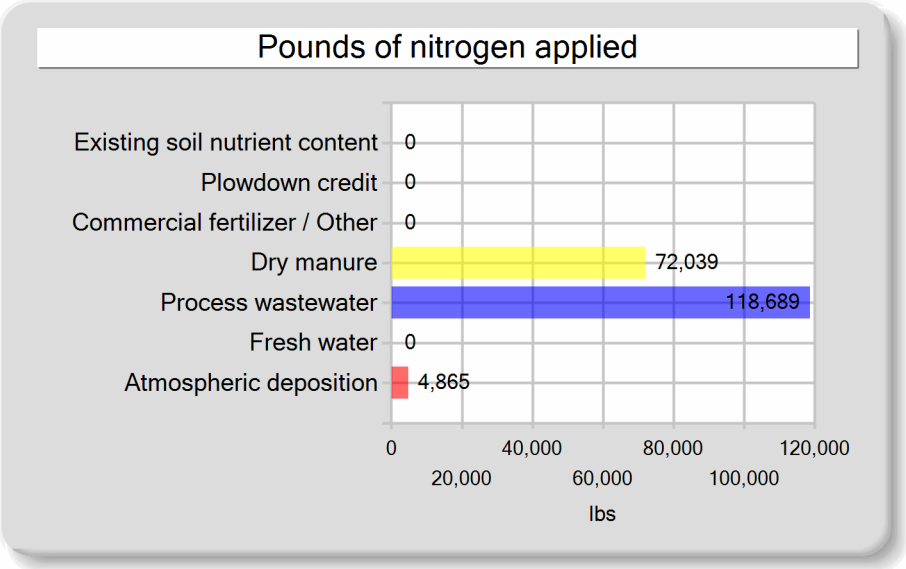
## A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

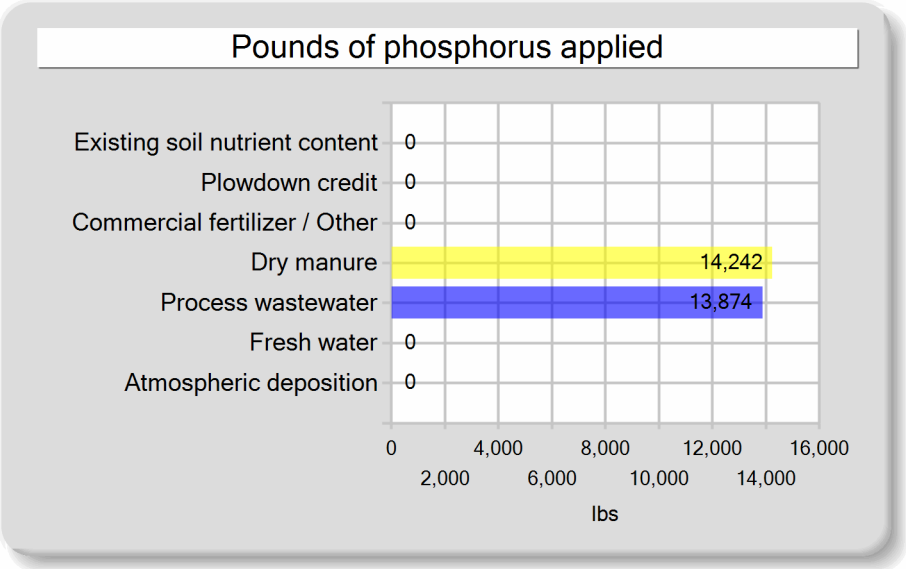
| | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 72,039.13 | 14,242.24 | 21,517.72 | 0.00 |
| Process wastewater | 118,689.36 | 13,873.63 | 168,413.42 | 962,780.99 |
| Fresh water | 0.00 | 0.00 | 0.00 | 138,638.48 |
| Atmospheric deposition | 4,865.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 195,593.49 | 28,115.87 | 189,931.13 | 1,101,419.47 |
| Anticipated crop nutrient removal | 139,392.00 | 24,523.80 | 114,920.40 | 0.00 |
| Actual crop nutrient removal | 146,454.97 | 29,136.14 | 199,058.77 | 657,207.60 |
| Nutrient balance | 49,138.52 | -1,020.27 | -9,127.63 | 444,211.87 |
| Applied to removed ratio | 1.34 | 0.96 | 0.95 | 1.68 |

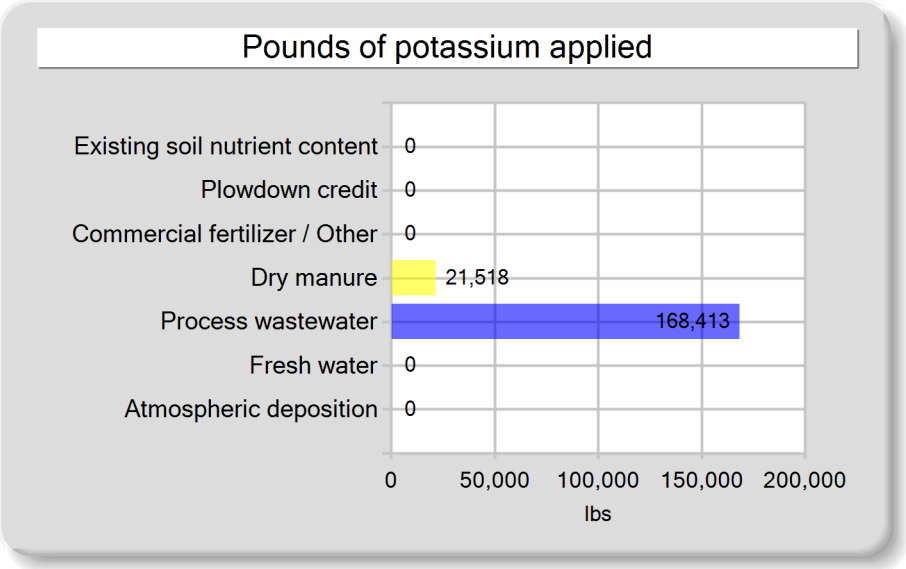
## B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

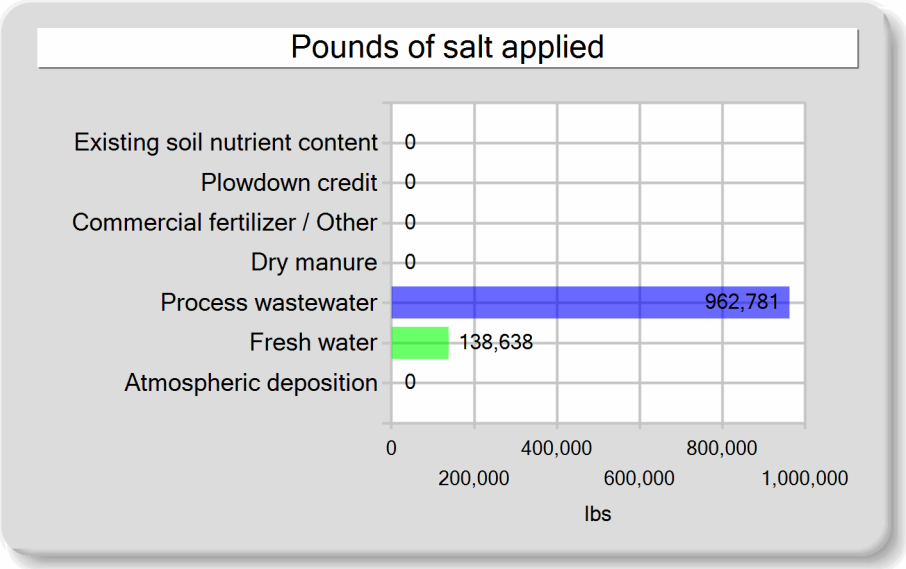


## C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE



## EXCEPTION REPORTING

### A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

*No manure or process wastewater discharges occurred during the reporting period.*

### B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

*No stormwater discharges occurred during the reporting period.*

### C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

*No land application area to surface water discharges occurred during the reporting period.*

## NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

### A. NUTRIENT MANAGEMENT PLAN STATEMENTS

| | |
|---|---|
| Was the facility's NMP updated in the reporting period? | Yes |
| Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |
| Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |

### B. EXPORT AGREEMENT STATEMENT

| | |
|---|---|
| Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period? | No |

ADDITIONAL NOTES

**A. NOTES**

**ALL SOIL PHOSPHORUS Results reported are expressed as pounds per acre of P2O5 or OSLEN***

CERTIFICATION

**A. OWNER AND/OR OPERATOR CERTIFICATION**

*I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.*

| | |
|---|---|
| SIGNATURE OF OWNER OF FACILITY | SIGNATURE OF OPERATOR OF FACILITY |
| Joe Oliveira | SAME AS OWNER |
| PRINT OR TYPE NAME | PRINT OR TYPE NAME |
| 06/21/2024 | |
| DATE | DATE |

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Joe Oliveira

Received: 08/22/2023 13:15
Reported: 08/28/2023 11:59

## Samples in this Report

| Lab ID | Sample | Matrix | Sampled By | Crop | Date Sampled |
|---|---|---|---|---|---|
| 23H1854-01 | Irrigation Pump | Ag Water | T. Madruga | | 08/21/2023 16:00 |

Default Cooler Temperature on Receipt °C: 0.1
Containers Intact
COC/Labels Agree
Preservation Confirmed
Received On Ice

## Notes and Definitions

| Item | Definition |
|---|---|
| H | Hold Time Exceeded |
| MCL | Drinking Water Maximum Contaminant Level |
| ND | Analyte NOT DETECTED at or above the reporting limit. |
| NES | Not Enough Sample |
| * | Not Taken |
| RPD | Relative Percent Difference |
| %REC | Percent Recovery |
| Source | Sample that was matrix spiked or duplicated. |

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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## Sample Results

**Sample: Irrigation Pump**
**23H1854-01 (Water)**

Sampled: 8/21/2023 16:00
Sampled By: T. Madruga

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **0.71** | mmhos/cm | 0.01 | 1 | | 08/23/23 13:42 | SM 2510 B | | BEH1083 |
| **Electrical Conductivity umhos** | **711** | umhos/cm | 10.0 | 1 | | 08/23/23 13:42 | SM 2510 B | | BEH1083 |
| **Nitrate Nitrogen as NO3N** | **23.9** | mg/L | 0.1 | 1 | 10 | 08/22/23 22:37 | EPA 300.0 | | BEH1061 |
| **pH** | **7.4** | units | 1.0 | 1 | | 08/23/23 13:42 | SM 4500-H+ | H | BEH1083 |
| **Total Filterable Solids (TDS)** | **545** | mg/L | 10.0 | 1 | | 08/24/23 09:12 | SM 2540 C | | BEH1058 |
| **Temperature** | **25.0** | °C | 0.0 | 1 | | 08/23/23 13:42 | SM 2510 B | | BEH1083 |
| Kjeldahl Nitrogen (TKN), Total | ND | mg/L | 1.00 | 1 | | 08/25/23 13:30 | SM 4500-NH3 C | | BEH1135 |
| **Total Nitrogen** | **23.9** | mg/L | 1.00 | 1 | | 08/25/23 13:30 | SM 4500-NH3 C | | BEH1135 |

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Reported: 08/28/2023 11:59

## Quality Control

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEH1058*** | | | | | | | | | | |
| **Blank (BEH1058-BLK1)** | | | | | Prepared: 8/22/2023 | Analyzed: 8/24/2023 | | | | |
| Total Filterable Solids (TDS) | ND | | 10.0 | mg/L | | | | | | |
| **LCS (BEH1058-BS1)** | | | | | Prepared: 8/22/2023 | Analyzed: 8/24/2023 | | | | |
| Total Filterable Solids (TDS) | 26.2 | | 10.0 | mg/L | 2000 | | 1.31 | 0-200 | | |
| **Duplicate (BEH1058-DUP1)** | | **Source: 23H1834-02** | | | Prepared: 8/22/2023 | Analyzed: 8/24/2023 | | | | |
| Total Filterable Solids (TDS) | 720 | | 10.0 | mg/L | | 710 | | | 1.40 | 10 |
| **Duplicate (BEH1058-DUP2)** | | **Source: 23H1845-01** | | | Prepared: 8/22/2023 | Analyzed: 8/24/2023 | | | | |
| Total Filterable Solids (TDS) | 140 | | 10.0 | mg/L | | 150 | | | 6.90 | 10 |
| **Reference (BEH1058-SRM1)** | | | | | Prepared: 8/22/2023 | Analyzed: 8/24/2023 | | | | |
| Total Filterable Solids (TDS) | 327 | | | mg/L | 325.0 | | 101 | 90-110 | | |
| **Reference (BEH1058-SRM2)** | | | | | Prepared: 8/22/2023 | Analyzed: 8/24/2023 | | | | |
| Total Filterable Solids (TDS) | 490 | | | mg/L | 495.0 | | 99.0 | 90-110 | | |

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## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEH1061*** | | | | | | | | | | |
| **Blank (BEH1061-BLK1)** | | | | | Prepared & Analyzed: 8/22/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BEH1061-BLK2)** | | | | | Prepared & Analyzed: 8/22/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BEH1061-BLK3)** | | | | | Prepared: 8/22/2023 Analyzed: 8/23/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **LCS (BEH1061-BS1)** | | | | | Prepared & Analyzed: 8/22/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.1 | | 0.1 | mg/L | 5.000 | | 102 | 90-110 | | |
| **LCS (BEH1061-BS2)** | | | | | Prepared: 8/22/2023 Analyzed: 8/23/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.0 | | 0.1 | mg/L | 5.000 | | 100 | 90-110 | | |
| **Duplicate (BEH1061-DUP1)** | | **Source: 23H1832-01** | | | Prepared & Analyzed: 8/22/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 0.03 | | 0.1 | mg/L | | 0.03 | | | 0.00 | 10 |
| **Duplicate (BEH1061-DUP2)** | | **Source: 23H1909-01** | | | Prepared: 8/22/2023 Analyzed: 8/23/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 0.5 | | 0.1 | mg/L | | 0.5 | | | 0.218 | 10 |
| **Matrix Spike (BEH1061-MS1)** | | **Source: 23H1832-01** | | | Prepared & Analyzed: 8/22/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.1 | | 0.1 | mg/L | 5.000 | 0.03 | 102 | 90-110 | | |
| **Matrix Spike (BEH1061-MS2)** | | **Source: 23H1909-01** | | | Prepared: 8/22/2023 Analyzed: 8/23/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.6 | | 0.1 | mg/L | 5.000 | 0.5 | 102 | 90-110 | | |
| **Reference (BEH1061-SRM1)** | | | | | Prepared & Analyzed: 8/22/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 9.9 | | | mg/L | 10.00 | | 99.1 | 90-110 | | |
| **Reference (BEH1061-SRM2)** | | | | | Prepared & Analyzed: 8/22/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 9.9 | | | mg/L | 10.00 | | 99.2 | 90-110 | | |
| **Reference (BEH1061-SRM3)** | | | | | Prepared: 8/22/2023 Analyzed: 8/23/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 9.9 | | | mg/L | 10.00 | | 99.2 | 90-110 | | |

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Reported: 08/28/2023 11:59

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEH1083*** | | | | | | | | | | |
| **Blank (BEH1083-BLK1)** | | | | | Prepared: 8/22/2023 | Analyzed: 8/23/2023 | | | | |
| pH | 5.5 | | 1.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Temperature | 25.0 | | 0.0 | °C | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| **Blank (BEH1083-BLK2)** | | | | | Prepared: 8/22/2023 | Analyzed: 8/23/2023 | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| pH | 6.0 | | 1.0 | units | | | | | | |
| Temperature | 25.0 | | 0.0 | °C | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| **Blank (BEH1083-BLK3)** | | | | | Prepared: 8/22/2023 | Analyzed: 8/23/2023 | | | | |
| pH | 6.0 | | 1.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Temperature | 25.0 | | 0.0 | °C | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| **Duplicate (BEH1083-DUP1)** | | **Source: 23H1832-05** | | | Prepared: 8/22/2023 | Analyzed: 8/23/2023 | | | | |
| pH | 8.6 | | 1.0 | units | | 8.6 | | | 0.00 | 10 |
| Electrical Conductivity | 1.39 | | 0.01 | mmhos/cm | | 1.39 | | | 0.122 | 10 |
| Electrical Conductivity umhos | 1390 | | 10.0 | umhos/cm | | 1390 | | | 0.122 | 10 |
| **Duplicate (BEH1083-DUP2)** | | **Source: 23H1832-09** | | | Prepared: 8/22/2023 | Analyzed: 8/23/2023 | | | | |
| Electrical Conductivity | 3.79 | | 0.01 | mmhos/cm | | 4.03 | | | 6.16 | 10 |
| pH | 8.2 | | 1.0 | units | | 8.2 | | | 0.366 | 10 |
| Electrical Conductivity umhos | 3790 | | 10.0 | umhos/cm | | 4030 | | | 6.16 | 10 |
| **Reference (BEH1083-SRM1)** | | | | | Prepared: 8/22/2023 | Analyzed: 8/23/2023 | | | | |
| Electrical Conductivity | 525 | | | umhos/cm | 538.0 | | 97.6 | 90-110 | | |
| **Reference (BEH1083-SRM2)** | | | | | Prepared: 8/22/2023 | Analyzed: 8/23/2023 | | | | |
| pH | 5.8 | | | units | 5.820 | | 101 | 28178-101.71 | | |
| **Reference (BEH1083-SRM3)** | | | | | Prepared: 8/22/2023 | Analyzed: 8/23/2023 | | | | |
| Electrical Conductivity | 986 | | | umhos/cm | 1000 | | 98.6 | 90-110 | | |
| Electrical Conductivity umhos | 986 | | | umhos/cm | 1000 | | 98.6 | 90-110 | | |
| **Reference (BEH1083-SRM4)** | | | | | Prepared: 8/22/2023 | Analyzed: 8/23/2023 | | | | |
| Electrical Conductivity | 982 | | | umhos/cm | 1000 | | 98.2 | 90-110 | | |
| Electrical Conductivity umhos | 982 | | | umhos/cm | 1000 | | 98.2 | 90-110 | | |
| **Reference (BEH1083-SRM5)** | | | | | Prepared: 8/22/2023 | Analyzed: 8/23/2023 | | | | |
| Electrical Conductivity | 968 | | | umhos/cm | 1000 | | 96.8 | 90-110 | | |

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Reported: 08/28/2023 11:59

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEH1083 (Continued)*** | | | | | | | | | | |
| **Reference (BEH1083-SRM5)** | | | | | Prepared: 8/22/2023 | Analyzed: 8/23/2023 | | | | |
| Electrical Conductivity umhos | 968 | | | umhos/cm | 1000 | | 96.8 | 90-110 | | |
| **Reference (BEH1083-SRM6)** | | | | | Prepared: 8/22/2023 | Analyzed: 8/23/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |
| **Reference (BEH1083-SRM7)** | | | | | Prepared: 8/22/2023 | Analyzed: 8/23/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 100 | 97.5-102.5 | | |
| **Reference (BEH1083-SRM8)** | | | | | Prepared: 8/22/2023 | Analyzed: 8/23/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 100 | 97.5-102.5 | | |

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## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEH1135*** | | | | | | | | | | |
| **Blank (BEH1135-BLK1)** | | | | | Prepared: 8/23/2023 | Analyzed: 8/25/2023 | | | | |
| Kjeldahl Nitrogen (TKN), Total | ND | | 1.00 | mg/L | | | | | | |
| Total Nitrogen | ND | | 1.00 | mg/L | | | | | | |
| **Blank (BEH1135-BLK2)** | | | | | Prepared: 8/23/2023 | Analyzed: 8/25/2023 | | | | |
| Kjeldahl Nitrogen (TKN), Total | ND | | 1.00 | mg/L | | | | | | |
| Total Nitrogen | ND | | 1.00 | mg/L | | | | | | |
| **LCS (BEH1135-BS1)** | | | | | Prepared: 8/23/2023 | Analyzed: 8/25/2023 | | | | |
| Kjeldahl Nitrogen (TKN), Total | 5.92 | | 1.00 | mg/L | 5.709 | | 104 | 90-110 | | |
| **LCS (BEH1135-BS2)** | | | | | Prepared: 8/23/2023 | Analyzed: 8/25/2023 | | | | |
| Kjeldahl Nitrogen (TKN), Total | 5.93 | | 1.00 | mg/L | 5.709 | | 104 | 90-110 | | |
| **Duplicate (BEH1135-DUP1)** | | **Source: 23H1826-09** | | | Prepared: 8/23/2023 | Analyzed: 8/25/2023 | | | | |
| Kjeldahl Nitrogen (TKN), Total | ND | | 1.40 | mg/L | | ND | | | | 10 |
| **Duplicate (BEH1135-DUP2)** | | **Source: 23H1829-03** | | | Prepared: 8/23/2023 | Analyzed: 8/25/2023 | | | | |
| Kjeldahl Nitrogen (TKN), Total | ND | | 1.40 | mg/L | | ND | | | | 10 |
| **Matrix Spike (BEH1135-MS1)** | | **Source: 23H1826-09** | | | Prepared: 8/23/2023 | Analyzed: 8/25/2023 | | | | |
| Kjeldahl Nitrogen (TKN), Total | 8.30 | | 1.40 | mg/L | 7.992 | ND | 104 | 90-110 | | |
| **Matrix Spike (BEH1135-MS2)** | | **Source: 23H1829-03** | | | Prepared: 8/23/2023 | Analyzed: 8/25/2023 | | | | |
| Kjeldahl Nitrogen (TKN), Total | 7.73 | | 1.40 | mg/L | 7.992 | ND | 96.7 | 90-110 | | |
| **Reference (BEH1135-SRM1)** | | | | | Prepared: 8/23/2023 | Analyzed: 8/25/2023 | | | | |
| Kjeldahl Nitrogen (TKN), Total | 23.3 | | | mg/L | 23.80 | | 97.9 | 90-110 | | |

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08/22/23 13:15 23H1854

JG

# DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples 1 No. of Bottles: 2

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
**[X] Ag Water** [ ] Groundwater [ ] Monitoring Well

Results Need By

**Name:** R & J Avila Inc
**Address:** 8119 Ellington Avenue
**City:** Hilmar **State: CA** **Zip:** 95324
**Telephone:** (209) 669-7919 **Fax:** (209) 634-4604
**Cell/Email:** 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
**COPY TO:**
**REQUESTED BY:** Rui & John Avila
**PROJECT** Joe Oliveira
**ID CROP:** N/A

[X] Copy of Chain [X] QA/QC Documents
**Sampled By:** T. Madruga

**Other:**

**Analysis and Bottles Required: (Please indicate Analysis)**

( ) DWW1: EC, pH, $NO_3$-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus $SO_4$, $CO_3$, $HCO_3$, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
**( X ) DCW1: EC, $NO_3$-N, TDS + TKN & TN**
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, $NO_3$-N, $NH_4$-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus $SO_4$, $CO_3$, $HCO_3$, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

| | Description of Samples | Date Sampled | Time Sampled | Rec'd Temp °C | Field $NH_4$-N |
|---|---|---|---|---|---|
| 1 | Irrigation Pump | 8/21/2023 | 1600 | 0.1 | |
| 2 | | | | | |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |
| 6 | | | | | |
| 7 | | | | | |
| 8 | | | | | |
| 9 | | | | | |
| 10 | | | | | |

**CHAIN OF CUSTODY**

| Carrier | Signature | Company | Received (Date/Time) | Relinquished (Date/Time) |
|---|---|---|---|---|
| First | John Avila | R & J Avila, Inc. | 8/21/2023 @ 1600 | 8/22/2023 @ 900 |
| Second | JLF | Dellavalle Laboratory, Inc. | 8/22/2023 @ 900 | 8-22-23 1:05 |
| Third | | | | |
| Fourth | | | 8/22 1325 | |

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal* ). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all [illegible] its, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

***Invoicing Information:*** ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

| Amt Paid | Rec By | Check # | Date |
|---|---|---|---|
| | | | |

**Signature**

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

08/22/23 13:15 23H1854

**Shipping Information:** Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

**Shipping Container:** Ice Chest ☒ None ☐ Box ☐

**Refrigerant:** Wet Ice: ☒ Blue Ice: ☐ None: ☐

| Container(s) Received | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Plastics:** | | | | | | | | | | |
| 100 mL sterile plastic $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 250 mL unpreserved (White) | | | | | | | | | | |
| 250 mL $HNO_3$ (Red) | | | | | | | | | | |
| 250 mL $H_2SO_4$ (Yellow) | 1 | | | | | | | | | |
| 500 mL unpreserved (White) | | | | | | | | | | |
| 500 mL $HNO_3$ (Red) | | | | | | | | | | |
| 500 mL $H_2SO_4$ (Yellow) | | | | | | | | | | |
| 1 L unpreserved (White) | 1 | | | | | | | | | |
| 1 L unpreserved (BOD) (Purple) | | | | | | | | | | |
| 1 L $HNO_3$ (Red) | | | | | | | | | | |
| 1 L $H_2SO_4$ (Yellow) | | | | | | | | | | |
| **VOA Vials in sets of 3:** | | | | | | | | | | |
| 40 mL VOA, HCl (Blue) | | | | | | | | | | |
| 40 mL VOA, $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 40 mL VOA, unpreserved (White) | | | | | | | | | | |
| **Glass:** | | | | | | | | | | |
| 250 mL AG unpreserved (White) | | | | | | | | | | |
| 250 mL AG $H_2SO_4$ (Yellow) | | | | | | | | | | |
| 250 mL AG $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 250 mL AG $Na_2S_2O_3$ + MCAA | | | | | | | | | | |
| 500 mL glass unpreserved (White) | | | | | | | | | | |
| 500 mL AG HCl (Blue) | | | | | | | | | | |
| 1 L AG unpreserved (White) | | | | | | | | | | |
| 1 L AG $H_2SO_4$ (Yellow) | | | | | | | | | | |
| 1 L AG $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 1 L AG HCl (Blue) | | | | | | | | | | |
| **Special:** | | | | | | | | | | |
| $Cr^{6+}$ w/Borate/HCO3/CO3 | | | | | | | | | | |
| Cyanide - 500 mL NaOH | | | | | | | | | | |
| Asbestos - 2 - 1L P wrapped in foil | | | | | | | | | | |
| 500 mL or 1 L **AP** $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 1 L **AG** or **P** NaOH + ZnAc (Sulfide) | | | | | | | | | | |
| Chlorite/Bromate 250 mL AG with EDA | | | | | | | | | | |
| **TTHM** | | | | | | | | | | |
| **HAA5** | | | | | | | | | | |
| 40 mL VOA, $Na_2S_2O_3$ + MCAA | | | | | | | | | | |

pH Strips
Lot 10BDH4501
Exp. Jan. 2025

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Joe Oliveria Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 11:00

## Samples in this Report

| Lab ID | Sample | Matrix | Sampled By | Crop | Date Sampled |
|---|---|---|---|---|---|
| 23K0899-01 | Domestic Well East | Ag Water | T. Madruga | | 11/15/2023 14:30 |
| 23K0899-02 | Domestic Well West | Ag Water | T. Madruga | | 11/15/2023 15:00 |

Default Cooler Temperature on Receipt °C: -1.1
Containers Intact
COC/Labels Agree
Received On Ice

## Notes and Definitions

| Item | Definition |
|---|---|
| H | Hold Time Exceeded |
| MCL | Drinking Water Maximum Contaminant Level |
| ND | Analyte NOT DETECTED at or above the reporting limit. |
| NES | Not Enough Sample |
| * | Not Taken |
| RPD | Relative Percent Difference |
| %REC | Percent Recovery |
| Source | Sample that was matrix spiked or duplicated. |

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Ranch: Joe Oliveria Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 11:00

## Sample Results

**Sample: Domestic Well East**
**23K0899-01 (Water)**

Sampled: 11/15/2023 14:30
Sampled By: T. Madruga

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **1.64** | mmhos/cm | 0.01 | 1 | | 11/20/23 14:02 | SM 2510 B | | BEK0616 |
| **Electrical Conductivity umhos** | **1640** | umhos/cm | 10.0 | 1 | | 11/20/23 14:02 | SM 2510 B | | BEK0616 |
| Ammonia (as N) | ND | mg/L | 0.00 | 1 | | 11/15/23 14:30 | Field | | BEK0559 |
| **Nitrate Nitrogen as NO3N** | **53.1** | mg/L | 0.1 | 1 | 10 | 11/16/23 21:32 | EPA 300.0 | | BEK0573 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 11/20/23 14:02 | SM 4500-H+ | H | BEK0616 |
| **pH** | **7.0** | units | 1.0 | 1 | | 11/20/23 14:02 | SM 4500-H+ | H | BEK0616 |

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## Sample Results
**(Continued)**

**Sample: Domestic Well West**
**23K0899-02 (Water)**

Sampled: 11/15/2023 15:00
Sampled By: T. Madruga

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **1.53** | mmhos/cm | 0.01 | 1 | | 11/20/23 14:16 | SM 2510 B | | BEK0617 |
| **Electrical Conductivity umhos** | **1530** | umhos/cm | 10.0 | 1 | | 11/20/23 14:16 | SM 2510 B | | BEK0617 |
| Ammonia (as N) | ND | mg/L | 0.00 | 1 | | 11/15/23 15:00 | Field | | BEK0559 |
| **Nitrate Nitrogen as NO3N** | **62.7** | mg/L | 0.1 | 1 | 10 | 11/16/23 21:52 | EPA 300.0 | | BEK0573 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 11/20/23 14:16 | SM 4500-H+ | H | BEK0617 |
| **pH** | **6.8** | units | 1.0 | 1 | | 11/20/23 14:16 | SM 4500-H+ | H | BEK0617 |

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Account# 00-0019840
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Ranch: Joe Oliveria Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 11:00

## Quality Control

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEK0573*** | | | | | | | | | | |
| **Blank (BEK0573-BLK1)** | | | | | Prepared & Analyzed: 11/16/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BEK0573-BLK2)** | | | | | Prepared & Analyzed: 11/16/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BEK0573-BLK3)** | | | | | Prepared: 11/16/2023 Analyzed: 11/17/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BEK0573-BLK4)** | | | | | Prepared: 11/16/2023 Analyzed: 11/17/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **LCS (BEK0573-BS1)** | | | | | Prepared & Analyzed: 11/16/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.2 | | 0.1 | mg/L | 5.000 | | 103 | 90-110 | | |
| **LCS (BEK0573-BS2)** | | | | | Prepared: 11/16/2023 Analyzed: 11/17/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.2 | | 0.1 | mg/L | 5.000 | | 104 | 90-110 | | |
| **LCS (BEK0573-BS3)** | | | | | Prepared: 11/16/2023 Analyzed: 11/17/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.2 | | 0.1 | mg/L | 5.000 | | 103 | 90-110 | | |
| **Duplicate (BEK0573-DUP1)** | | **Source: 23K0896-02** | | | Prepared & Analyzed: 11/16/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.8 | | 0.1 | mg/L | | 5.8 | | | 0.173 | 10 |
| **Duplicate (BEK0573-DUP2)** | | **Source: 23K0901-01** | | | Prepared: 11/16/2023 Analyzed: 11/17/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 0.3 | | 0.1 | mg/L | | 0.3 | | | 0.629 | 10 |
| **Duplicate (BEK0573-DUP3)** | | **Source: 23K0935-02** | | | Prepared: 11/16/2023 Analyzed: 11/17/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 1.5 | | 0.1 | mg/L | | 1.5 | | | 0.195 | 10 |
| **Matrix Spike (BEK0573-MS1)** | | **Source: 23K0896-02** | | | Prepared & Analyzed: 11/16/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 11.0 | | 0.1 | mg/L | 5.000 | 5.8 | 104 | 90-110 | | |
| **Matrix Spike (BEK0573-MS2)** | | **Source: 23K0901-01** | | | Prepared: 11/16/2023 Analyzed: 11/17/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.2 | | 0.1 | mg/L | 5.000 | 0.3 | 97.1 | 90-110 | | |
| **Matrix Spike (BEK0573-MS3)** | | **Source: 23K0935-02** | | | Prepared: 11/16/2023 Analyzed: 11/17/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 6.8 | | 0.1 | mg/L | 5.000 | 1.5 | 106 | 90-110 | | |
| **Reference (BEK0573-SRM1)** | | | | | Prepared & Analyzed: 11/16/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 10.2 | | | mg/L | 10.00 | | 102 | 90-110 | | |
| **Reference (BEK0573-SRM2)** | | | | | Prepared: 11/16/2023 Analyzed: 11/17/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 10.3 | | | mg/L | 10.00 | | 103 | 90-110 | | |

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Ranch: Joe Oliveria Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 11:00

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEK0573 (Continued)*** | | | | | | | | | | |
| **Reference (BEK0573-SRM3)** | | | | | Prepared: 11/16/2023 | Analyzed: 11/17/2023 | | | | |
| Nitrate Nitrogen as NO3N | 10.3 | | | mg/L | 10.00 | | 103 | 90-110 | | |
| **Reference (BEK0573-SRM4)** | | | | | Prepared: 11/16/2023 | Analyzed: 11/17/2023 | | | | |
| Nitrate Nitrogen as NO3N | 10.3 | | | mg/L | 10.00 | | 103 | 90-110 | | |

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Ranch: Joe Oliveria Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 11:00

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEK0616*** | | | | | | | | | | |
| **Blank (BEK0616-BLK1)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 5.7 | | 1.0 | units | | | | | | |
| **Blank (BEK0616-BLK2)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 6.2 | | 1.0 | units | | | | | | |
| **Blank (BEK0616-BLK3)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 6.7 | | 1.0 | units | | | | | | |
| **Duplicate (BEK0616-DUP1)** | | **Source: 23K0896-03** | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 0.47 | | 0.01 | mmhos/cm | | 0.47 | | | 0.423 | 10 |
| pH | 7.8 | | 1.0 | units | | 7.8 | | | 0.255 | 10 |
| Electrical Conductivity umhos | 472 | | 10.0 | umhos/cm | | 474 | | | 0.423 | 10 |
| **Duplicate (BEK0616-DUP2)** | | **Source: 23K0899-01** | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 1.65 | | 0.01 | mmhos/cm | | 1.64 | | | 0.298 | 10 |
| Electrical Conductivity umhos | 1650 | | 10.0 | umhos/cm | | 1640 | | | 0.298 | 10 |
| pH | 7.0 | | 1.0 | units | | 7.0 | | | 0.00 | 10 |
| **Reference (BEK0616-SRM1)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 561 | | | umhos/cm | 538.0 | | 104 | 90-110 | | |
| **Reference (BEK0616-SRM2)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| pH | 7.6 | | | units | 7.520 | | 101 | 67021-101.32 | | |
| **Reference (BEK0616-SRM3)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 1040 | | | umhos/cm | 1000 | | 104 | 90-110 | | |
| Electrical Conductivity umhos | 1040 | | | umhos/cm | 1000 | | 104 | 90-110 | | |
| **Reference (BEK0616-SRM4)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 1020 | | | umhos/cm | 1000 | | 102 | 90-110 | | |
| Electrical Conductivity umhos | 1020 | | | umhos/cm | 1000 | | 102 | 90-110 | | |
| **Reference (BEK0616-SRM5)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 1030 | | | umhos/cm | 1000 | | 103 | 90-110 | | |

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Joe Oliveria Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 11:00

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEK0616 (Continued)*** | | | | | | | | | | |
| **Reference (BEK0616-SRM5)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity umhos | 1030 | | | umhos/cm | 1000 | | 103 | 90-110 | | |
| **Reference (BEK0616-SRM6)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |
| **Reference (BEK0616-SRM7)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |
| **Reference (BEK0616-SRM8)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 100 | 97.5-102.5 | | |

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Received: 11/16/2023 15:35
Reported: 11/22/2023 11:00

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEK0617*** | | | | | | | | | | |
| **Blank (BEK0617-BLK1)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 5.7 | | 1.0 | units | | | | | | |
| **Blank (BEK0617-BLK2)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| pH | 6.6 | | 1.0 | units | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| **Blank (BEK0617-BLK3)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| pH | 6.0 | | 1.0 | units | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| **Duplicate (BEK0617-DUP1)** | | **Source: 23K0902-03** | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 0.79 | | 0.01 | mmhos/cm | | 0.78 | | | 0.800 | 10 |
| pH | 7.7 | | 1.0 | units | | 7.7 | | | 0.130 | 10 |
| Electrical Conductivity umhos | 791 | | 10.0 | umhos/cm | | 785 | | | 0.800 | 10 |
| **Duplicate (BEK0617-DUP2)** | | **Source: 23K0904-01** | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 1.59 | | 0.01 | mmhos/cm | | 1.59 | | | 0.00 | 10 |
| pH | 8.0 | | 1.0 | units | | 8.0 | | | 0.125 | 10 |
| Electrical Conductivity umhos | 1590 | | 10.0 | umhos/cm | | 1590 | | | 0.00 | 10 |
| **Reference (BEK0617-SRM1)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 555 | | | umhos/cm | 538.0 | | 103 | 90-110 | | |
| **Reference (BEK0617-SRM2)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| pH | 7.6 | | | units | 7.520 | | 101 | 67021-101.32 | | |
| **Reference (BEK0617-SRM3)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 1030 | | | umhos/cm | 1000 | | 103 | 90-110 | | |
| Electrical Conductivity umhos | 1030 | | | umhos/cm | 1000 | | 103 | 90-110 | | |
| **Reference (BEK0617-SRM4)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 1040 | | | umhos/cm | 1000 | | 104 | 90-110 | | |
| Electrical Conductivity umhos | 1040 | | | umhos/cm | 1000 | | 104 | 90-110 | | |
| **Reference (BEK0617-SRM5)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity | 1040 | | | umhos/cm | 1000 | | 104 | 90-110 | | |

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Account# 00-0019840
Account Manager: Ben Nydam
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Received: 11/16/2023 15:35
Reported: 11/22/2023 11:00

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEK0617 (Continued)*** | | | | | | | | | | |
| **Reference (BEK0617-SRM5)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| Electrical Conductivity umhos | 1040 | | | umhos/cm | 1000 | | 104 | 90-110 | | |
| **Reference (BEK0617-SRM6)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 100 | 97.5-102.5 | | |
| **Reference (BEK0617-SRM7)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |
| **Reference (BEK0617-SRM8)** | | | | | Prepared: 11/17/2023 | Analyzed: 11/20/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |

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11/16/23 15:35 23K0899

# DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 2 No. of Bottles: 2

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Results Need By

**Name:** R & J Avila Inc
**Address:** 8119 Ellington Avenue
**City:** Hilmar **State: CA** **Zip:** 95324
**Telephone:** (209) 669-7919 **Fax:** (209) 634-4604
**Cell/Email:** 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
**COPY TO:**

**REQUESTED BY:** Rui & John Avila
**PROJECT** Joe Oliveira Dairy
**ID CROP:** N/A

[X] Copy of Chain [X] QA/QC Documents
**Sampled By:** T. Madruga

**Other:**

**Analysis and Bottles Required: (Please indicate Analysis)**

**( X ) DWW1: EC, pH, $NO_3$-N (NH4-N Field Test)**
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus $SO_4$, $CO_3$, $HCO_3$, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, $NO_3$-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, $NO_3$-N, $NH_4$-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus $SO_4$, $CO_3$, $HCO_3$, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

| | Description of Samples | Date Sampled | Time Sampled | Rec'd Temp °C | Field $NH_4$-N |
|---|---|---|---|---|---|
| 1 | Domestic Well East | 11/15/2023 | 1430 | -1.1 | ND |
| 2 | Domestic Well West | 11/15/2023 | 1500 | 2.1 | ND |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |
| 6 | | | | | |
| 7 | IR Thermometer SN: 221314357 | | | | |
| 8 | Correction Factor: 0°C | | | | |
| 9 | Calibration Due: 12/22/2023 | | | | |
| 10 | Location: Laboratory | | | | |

**CHAIN OF CUSTODY**

| Carrier | Signature | Company | Received (Date/Time) | Relinquished (Date/Time) |
|---|---|---|---|---|
| First | John Avila | R & J Avila, Inc. | 11/15/2023 @ 1500 | 11/16/2023 @9am |
| Second | JLF | Dellavalle Laboratory, Inc. | 11/16/2023 @9am | 11-16-23 3:25 |
| Third | | DLL | 11-16-23 15:35 | |
| Fourth | | | | |

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal* ). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

***Invoicing Information:*** ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

| Amt Paid | Rec By | Check # | Date |
|---|---|---|---|
| | | | |

**Signature**

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/16/23 15:35 23K0899

**Shipping Information:** Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

**Shipping Container:** Ice Chest ☐ None ☑ Box ☐

**Refrigerant:** Wet Ice:☑ Blue Ice:☐ None:☐

| **Container(s) Received** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Plastics:** | | | | | | | | | | |
| 100 mL sterile plastic $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 250 mL unpreserved (White) | | | | | | | | | | |
| 250 mL $HNO_3$ (Red) | | | | | | | | | | |
| 250 mL $H_2SO_4$ (Yellow) | | | | | | | | | | |
| 500 mL unpreserved (White) | | | | | | | | | | |
| 500 mL $HNO_3$ (Red) | | | | | | | | | | |
| 500 mL $H_2SO_4$ (Yellow) | | | | | | | | | | |
| 1 L unpreserved (White) | 1 | 1 | | | | | | | | |
| 1 L unpreserved (BOD) (Purple) | | | | | | | | | | |
| 1 L $HNO_3$ (Red) | | | | | | | | | | |
| 1 L $H_2SO_4$ (Yellow) | | | | | | | | | | |
| **VOA Vials in sets of 3:** | | | | | | | | | | |
| 40 mL VOA, HCl (Blue) | | | | | | | | | | |
| 40 mL VOA, $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 40 mL VOA, unpreserved (White) | | | | | | | | | | |
| **Glass:** | | | | | | | | | | |
| 250 mL AG unpreserved (White) | | | | | | | | | | |
| 250 mL AG $H_2SO_4$ (Yellow) | | | | | | | | | | |
| 250 mL AG $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 250 mL AG $Na_2S_2O_3$ + MCAA | | | | | | | | | | |
| 500 mL glass unpreserved (White) | | | | | | | | | | |
| 500 mL AG HCl (Blue) | | | | | | | | | | |
| 1 L AG unpreserved (White) | | | | | | | | | | |
| 1 L AG $H_2SO_4$ (Yellow) | | | | | | | | | | |
| 1 L AG $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 1 L AG HCl (Blue) | | | | | | | | | | |
| **Special:** | | | | | | | | | | |
| $Cr^{6+}$ w/Borate/HCO3/CO3 | | | | | | | | | | |
| Cyanide - 500 mL NaOH | | | | | | | | | | |
| Asbestos - 2 - 1L P wrapped in foil | | | | | | | | | | |
| 500 mL or 1 L **AP** $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 1 L **AG** or **P** NaOH + ZnAc (Sulfide) | | | | | | | | | | |
| Chlorite/Bromate 250 mL AG with EDA | | | | | | | | | | |
| **TTHM** | | | | | | | | | | |
| **HAA5** | | | | | | | | | | |
| 40 mL VOA, $Na_2S_2O_3$ + MCAA | | | | | | | | | | |